

COURSE GEST-H-502

SUPPLY CHAIN PERFORMANCE ANALYTICS

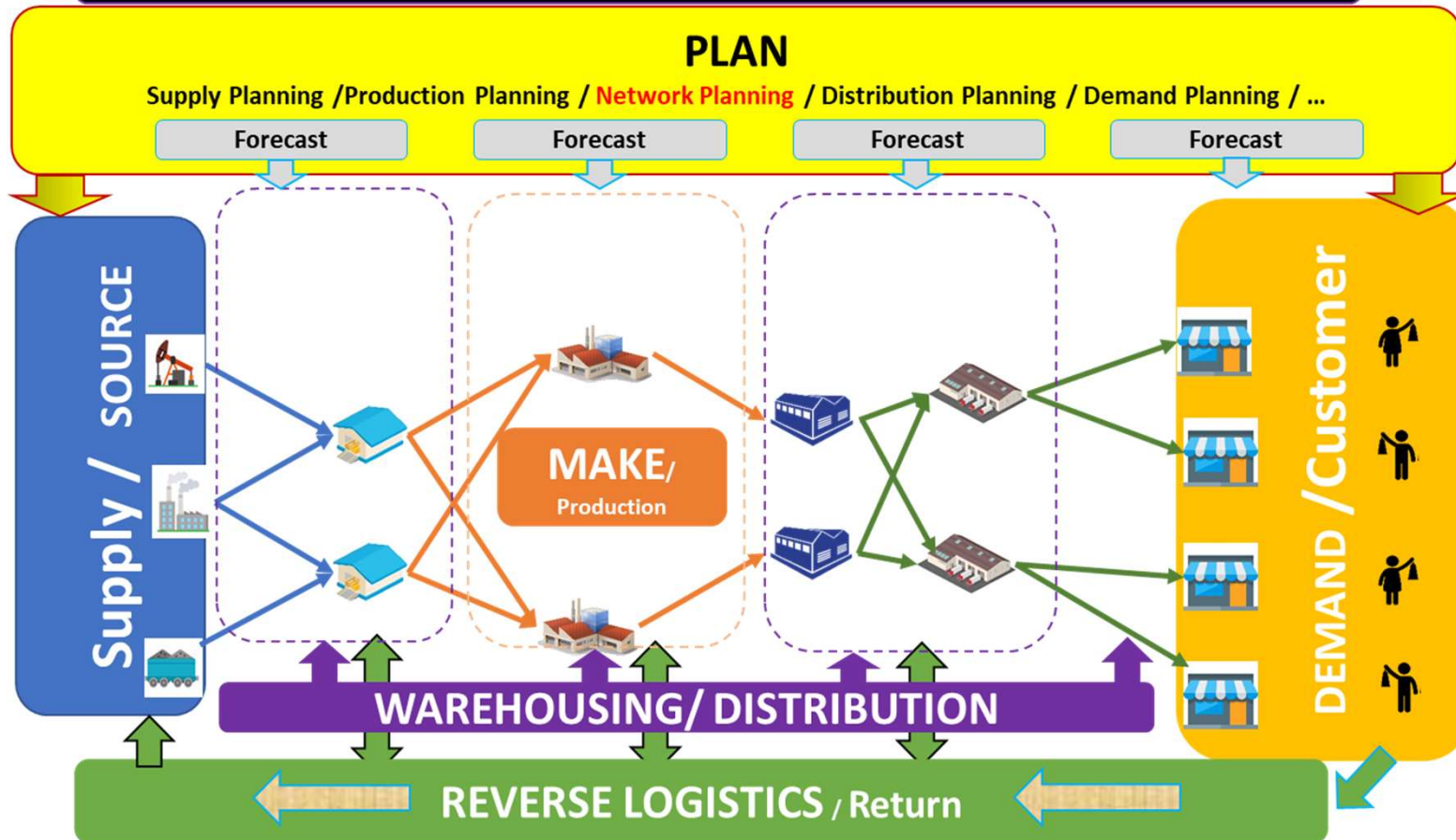
SUPPLY CHAIN CONCEPT AND SUPPLY CHAIN PERFORMANCE MANAGEMENT

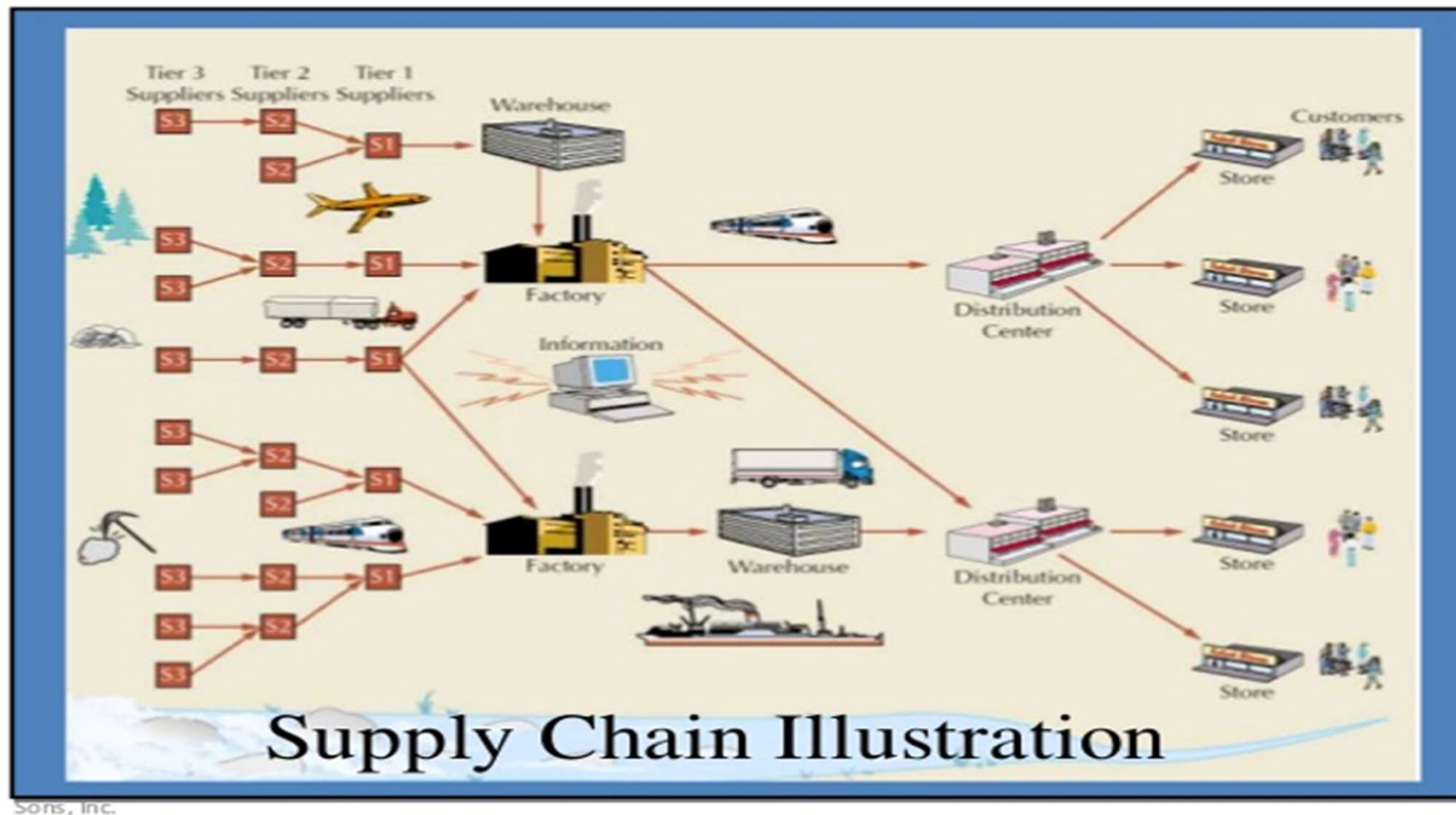
Prof. Dr. Alassane B. NDIAYE

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 - F1. Performance Management Science as an assessment framework*
 - F2. Building of a Supply Chain Performance Management Model*

MAIN BLOCS OF LOGISTICS OPERATIONS





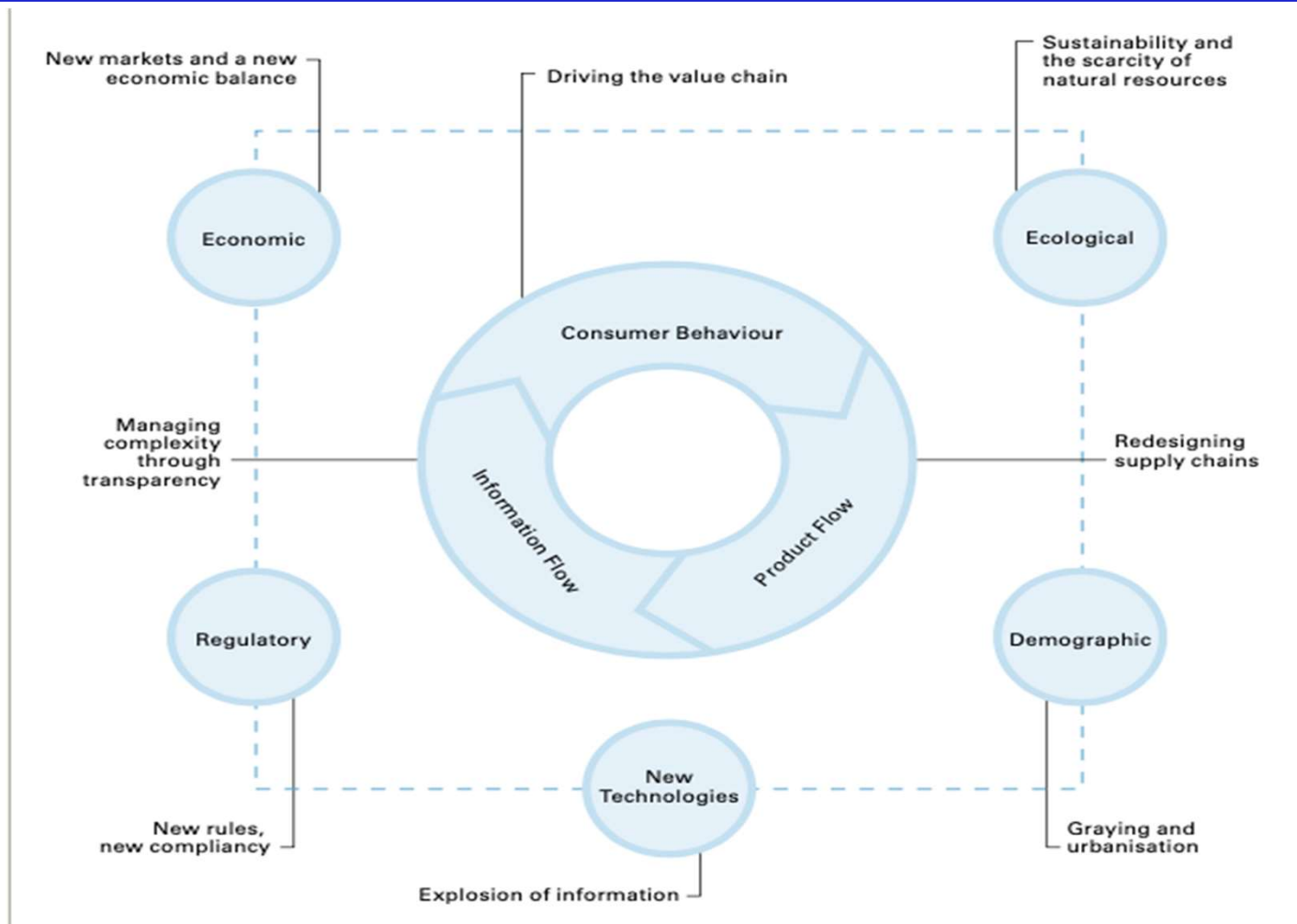
Sons, Inc.

The concept of Supply Chain Management:

Conception and pilotage of an integrated and coordinated system
(of materials, information, processes, functions and resources, etc.)
where the priority is given to the overall optimal performance and
not to the local optima .



Forces and Trends



Capgemini, 2016

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D6. The ENVIRONMENTALLY CLEAN/the GREEN addition

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D8. The SOCIALLY AND ETHICALLY RESPONSIBLE addition

D9. The LEAN addition

D10. The SMART / DIGITAL addition

E. **ELABORATING A STRATEGIC SUPPLY CHAIN MANAGEMENT STRATEGY**

E1. Which Strategic Supply Chain Management Strategy is right?

E2. A particular Strategic Supply Chain Management Strategy: The TRIPLE-A Strategy

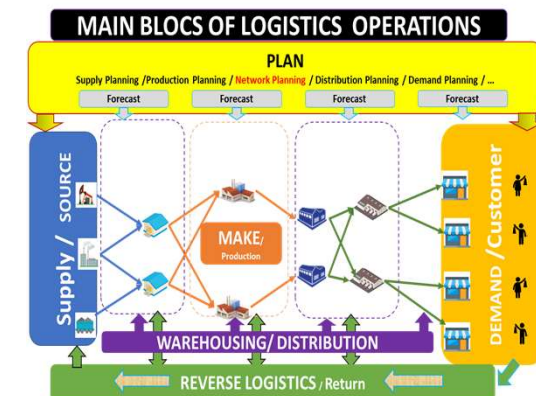
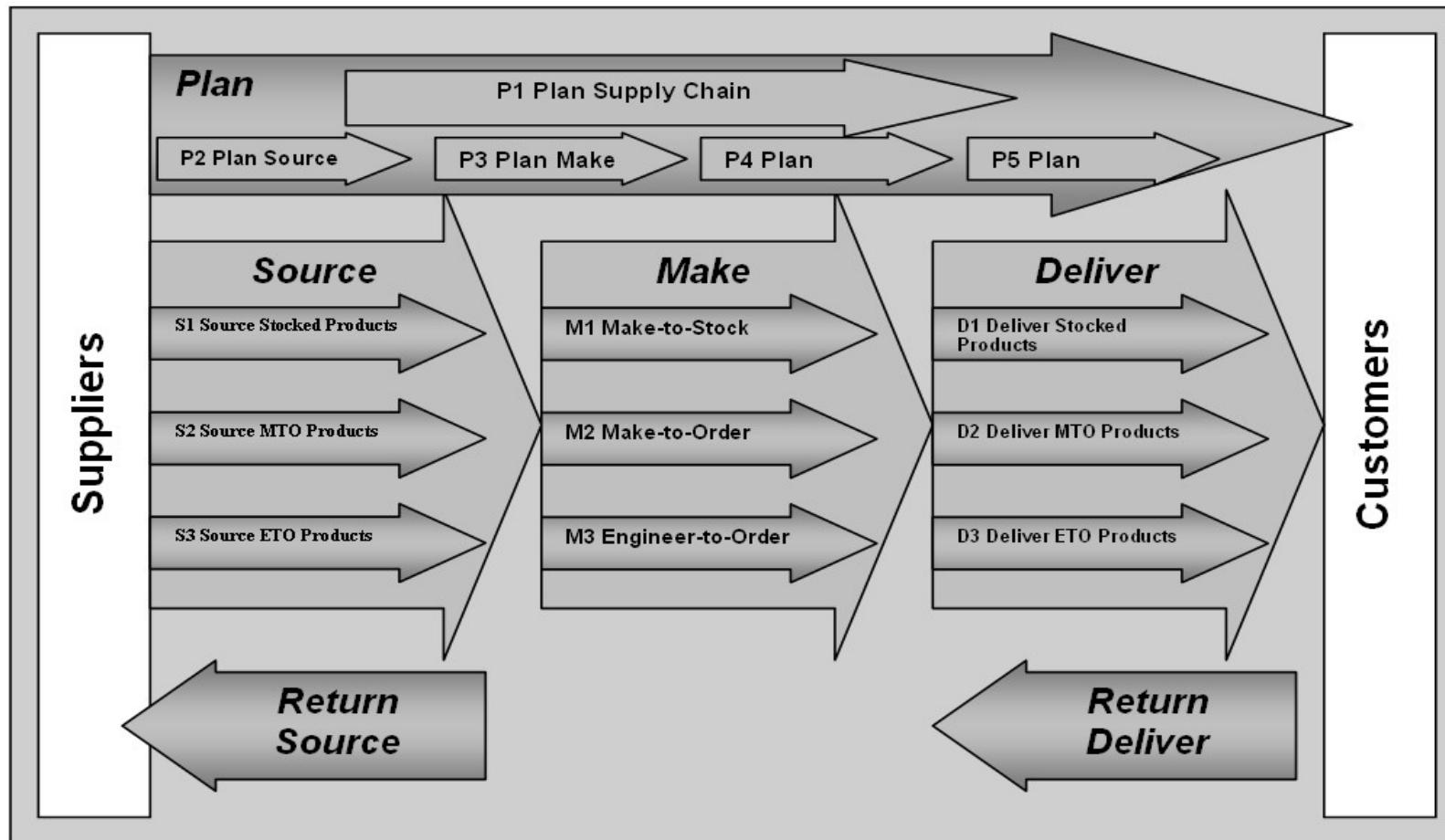
E3. Discussing the trade-offs paradigms in elaborating a Strategic Supply Chain Management Strategy

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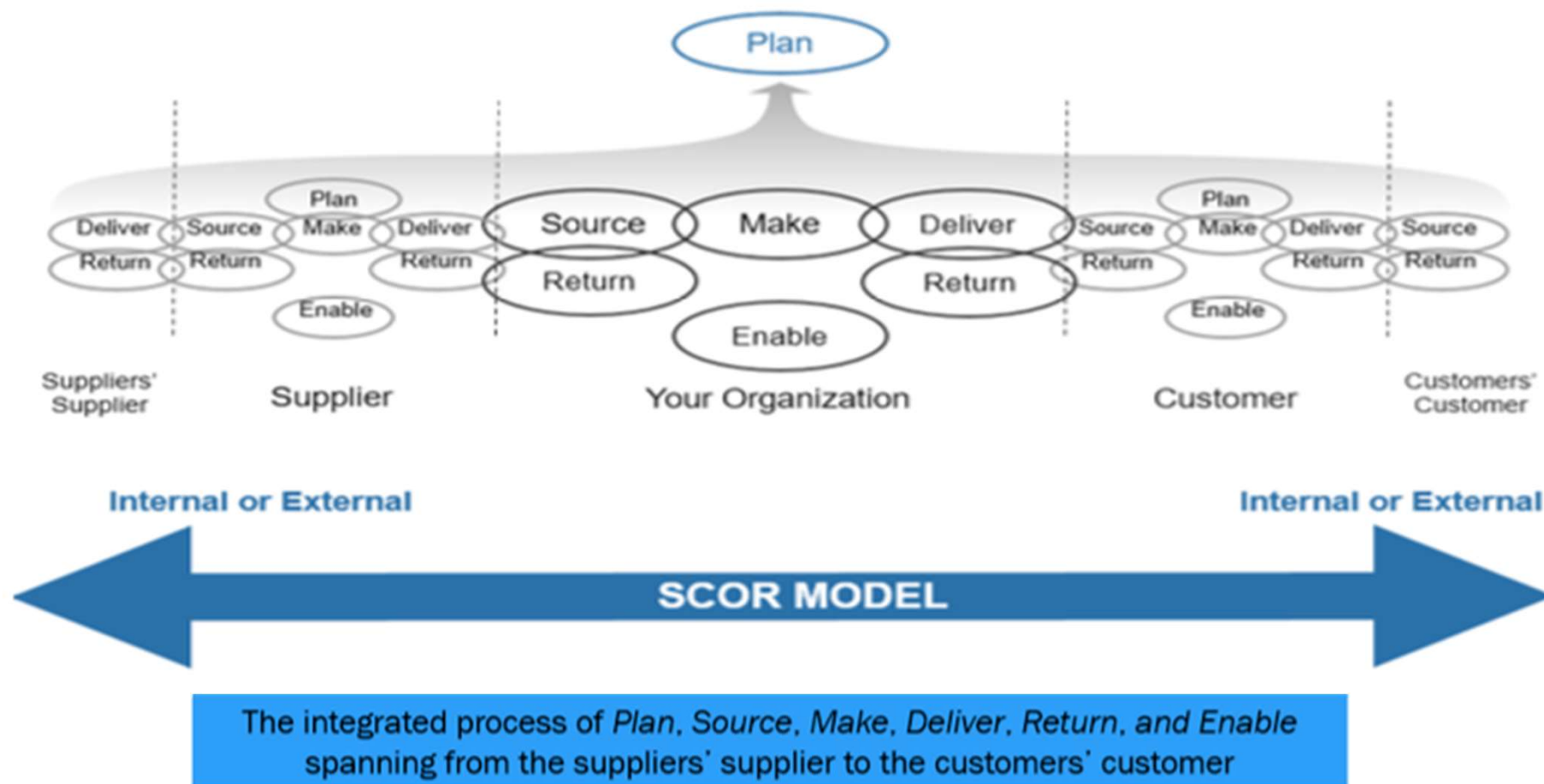
F1. Performance Management Science as an assessment framework

F2. Building of a Supply Chain Performance Management Model

SCOR®: A REFERENCE MODEL, A CROSS-INDUSTRY OPEN STANDARD

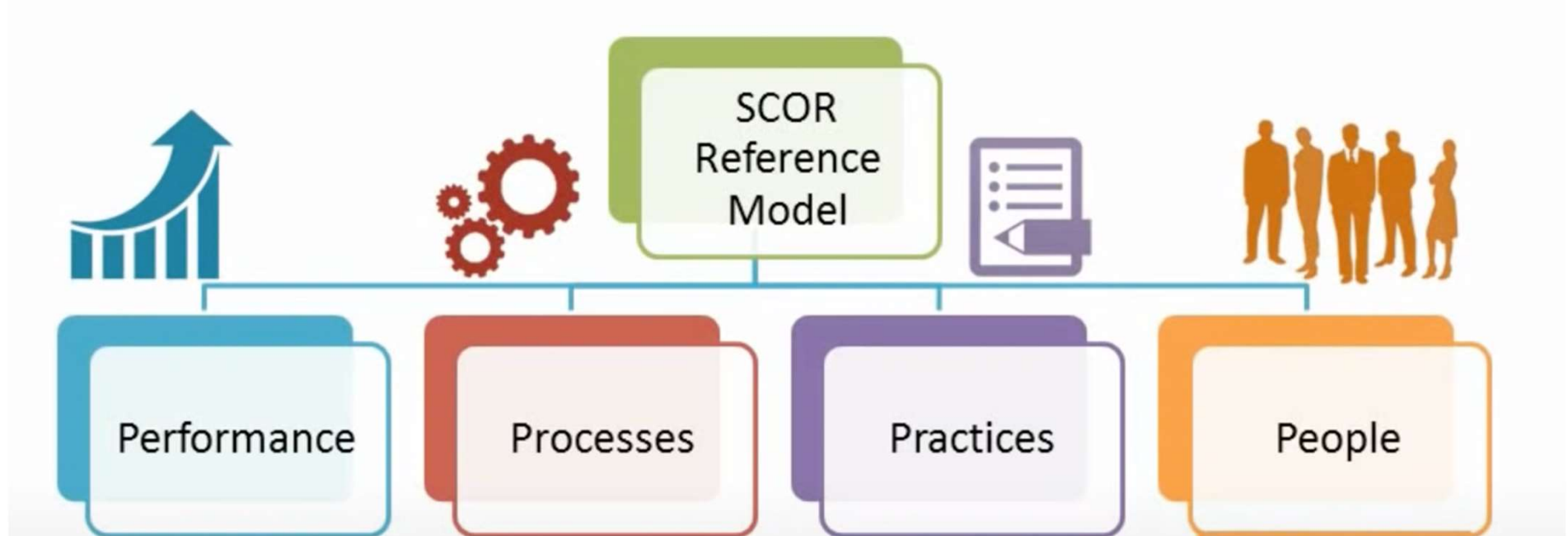


SCOR®: A REFERENCE MODEL, A CROSS-INDUSTRY OPEN STANDARD



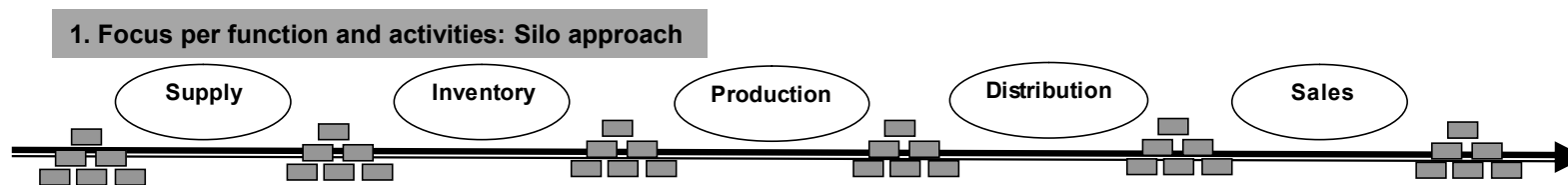
Process is a set of operations

Four Major Components



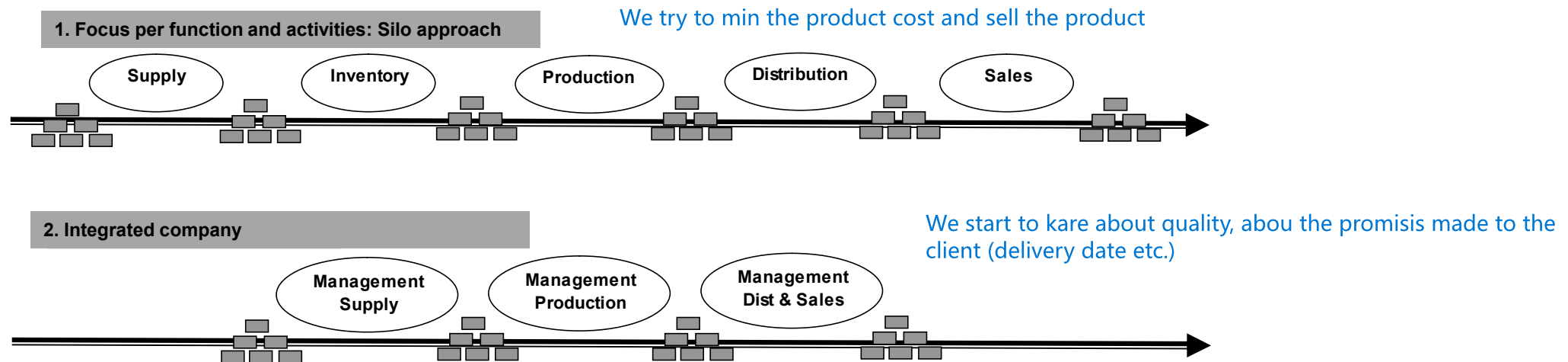
STAGE 1:

- Clustered « Vertical functions » and activities (Supply, Production, Sales,...)
- Target parameters: product cost



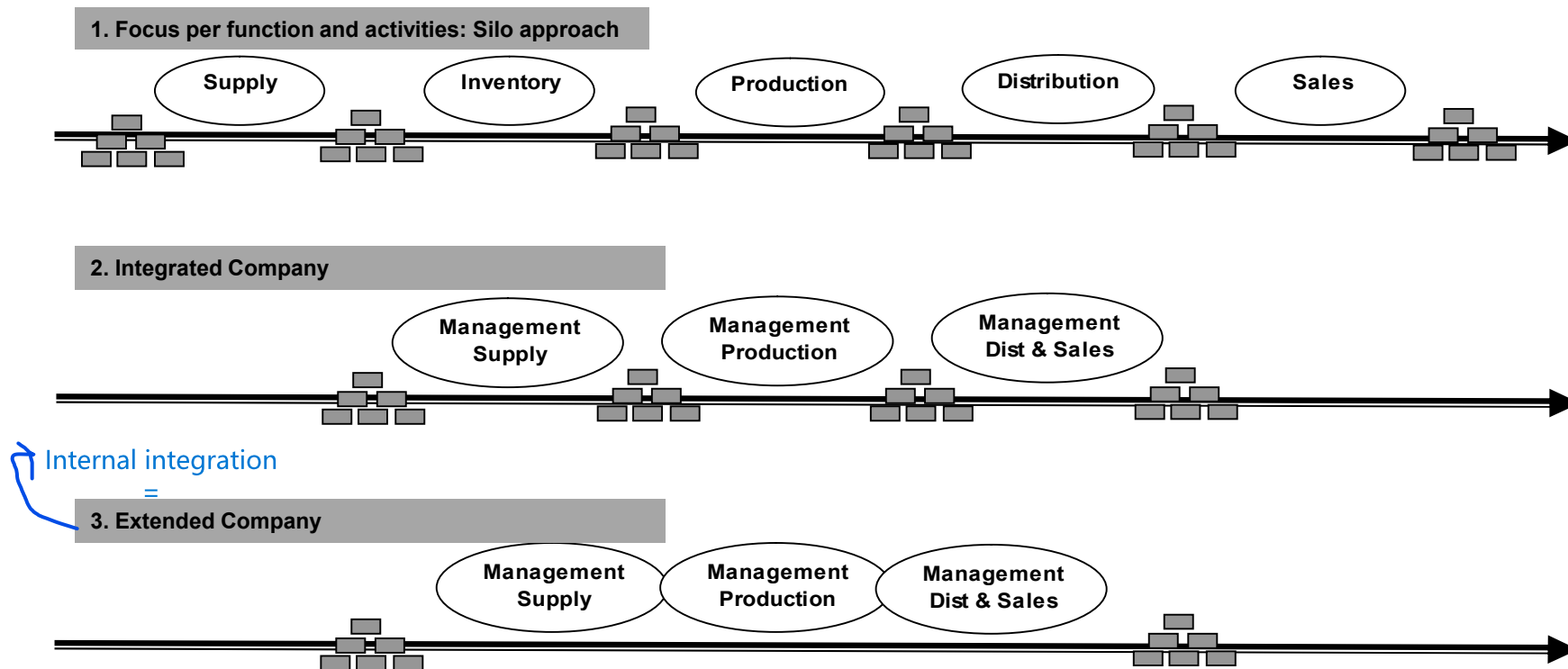
STAGE 2:

- Still clustered but some functions/activities are grouped and managed per groups
- Target parameters: product cost and product quality



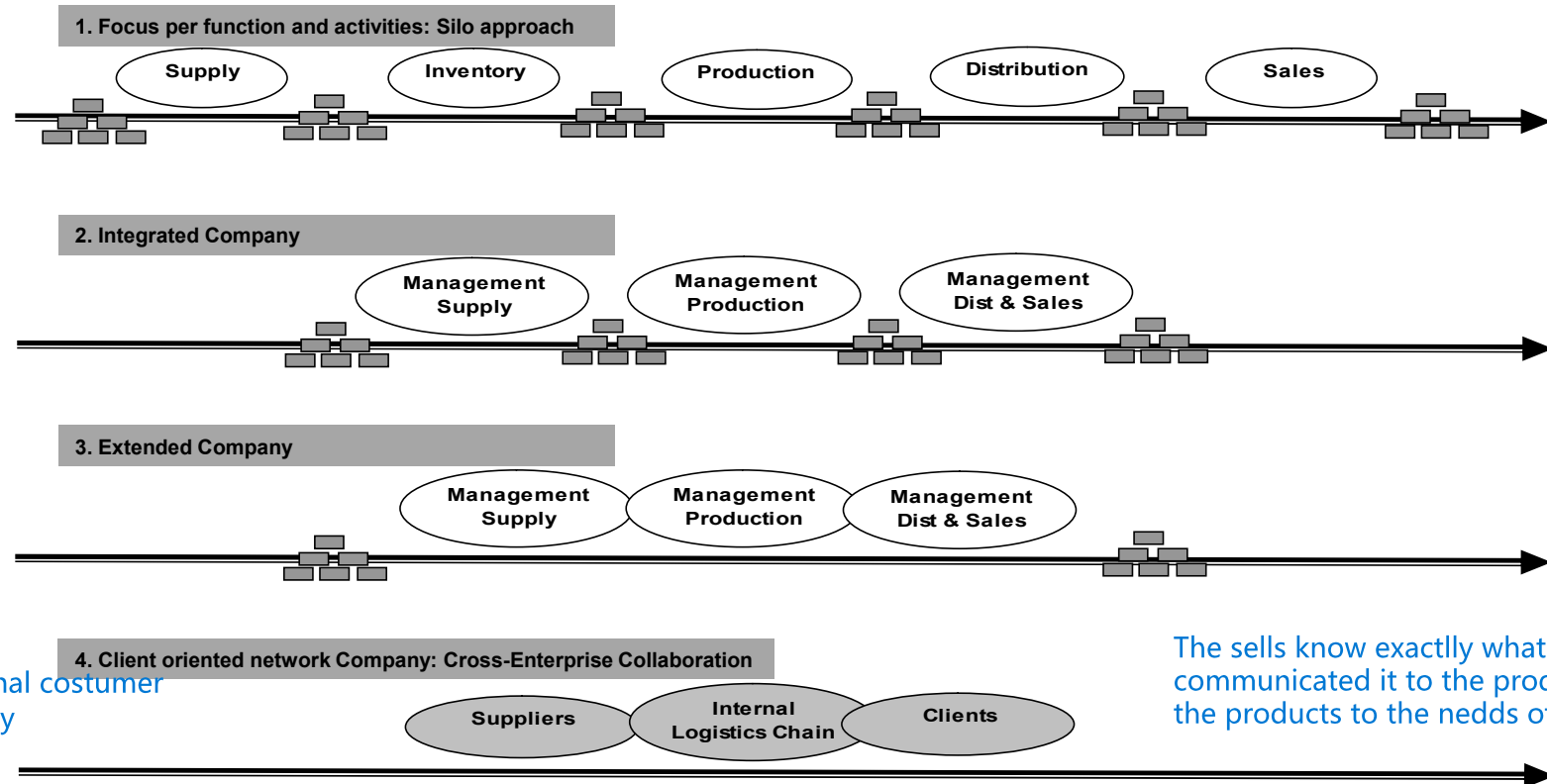
STAGE 3:

- « Extended Company»: Stage 2 + Cooperation with Suppliers & Distributors
- Target parameters: product cost, product quality and respect of made to the client



STAGE 4:

- « Client oriented Network »: Full integration and client-oriented + Supply Chain Manager
- Target parameters: Build a Competitive Advantage

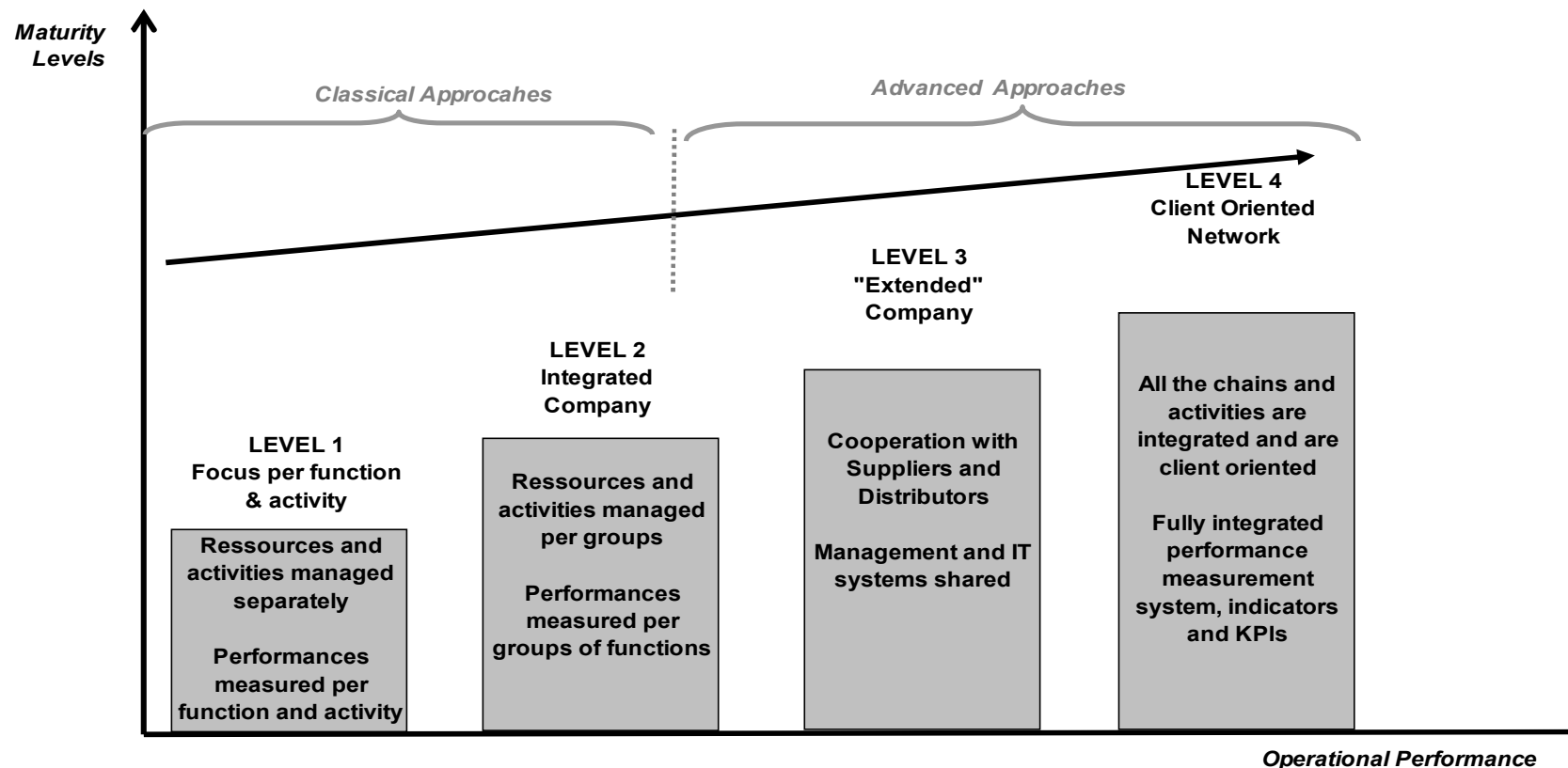


It's from the final customer to the company

The sells know exactly what are we buing so they can communicated it to the producer so the producer will tailor the products to the nedds of the clients.

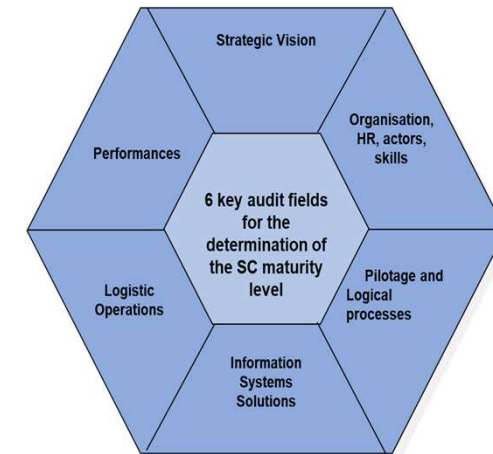
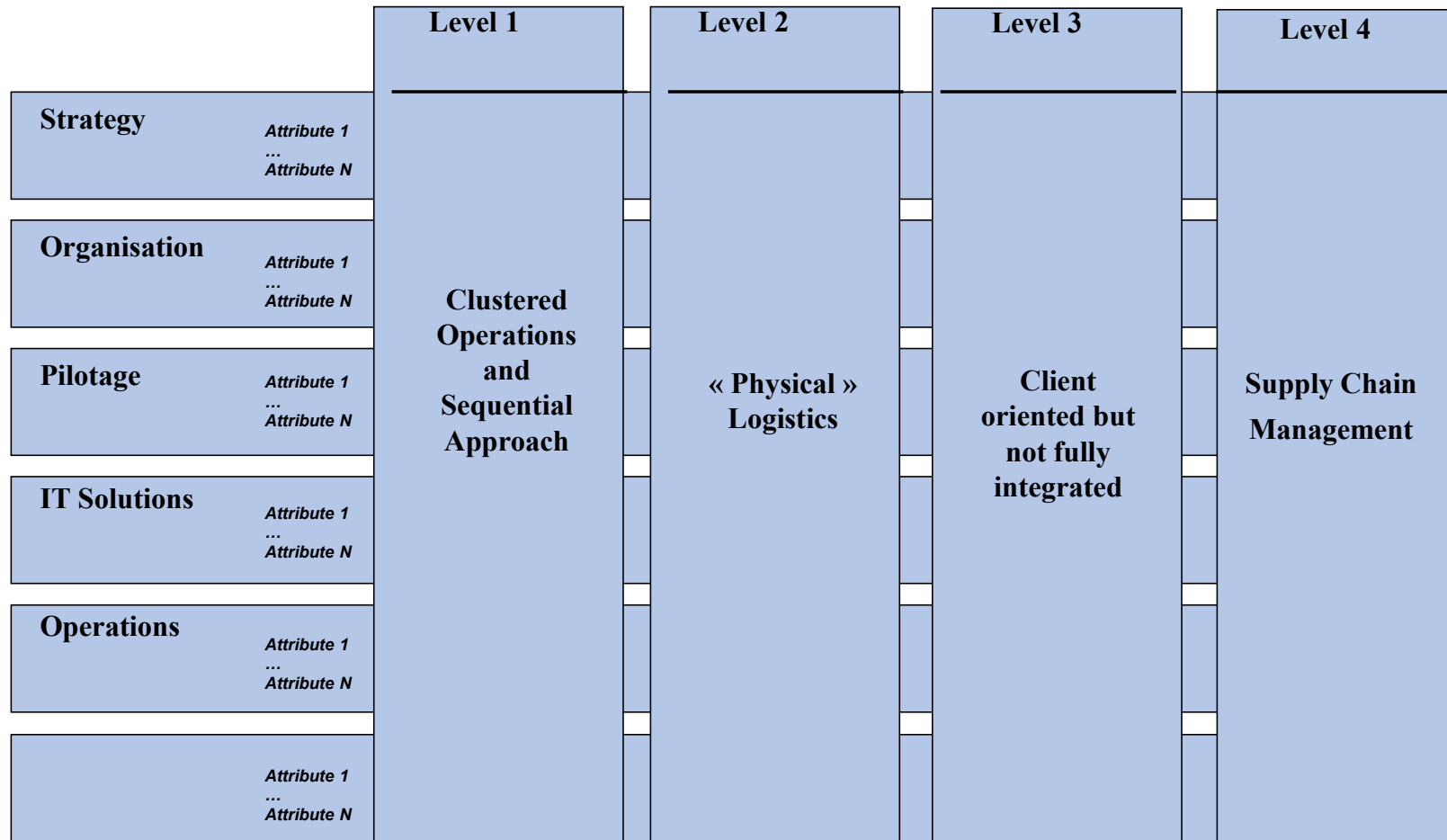
SUPPLY-CHAIN INTEGRATION

→ 4 LEVELS OF MATURITY CORRESPONDING TO 4 TYPES OF SUPPLY CHAIN INTEGRATION



B. SUPPLY CHAIN REFERENCE MODEL AND SUPPLY CHAIN INTEGRATION

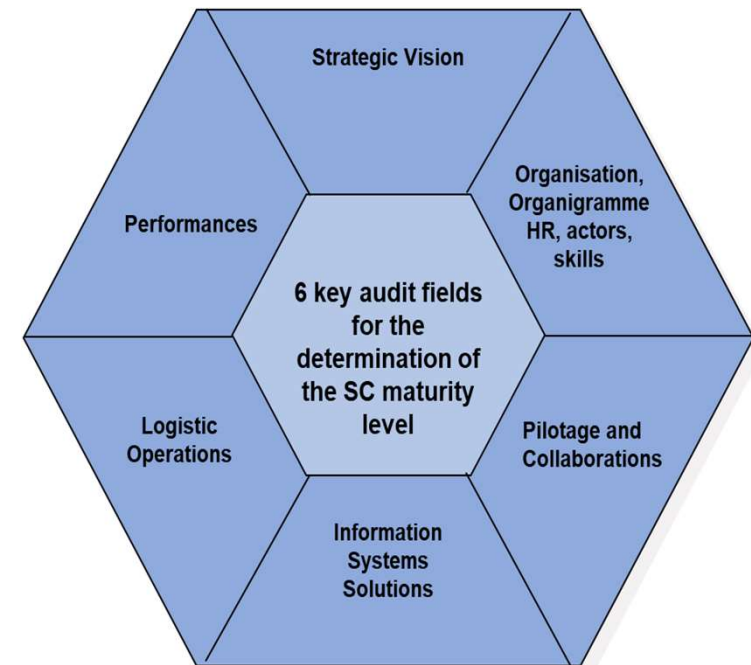
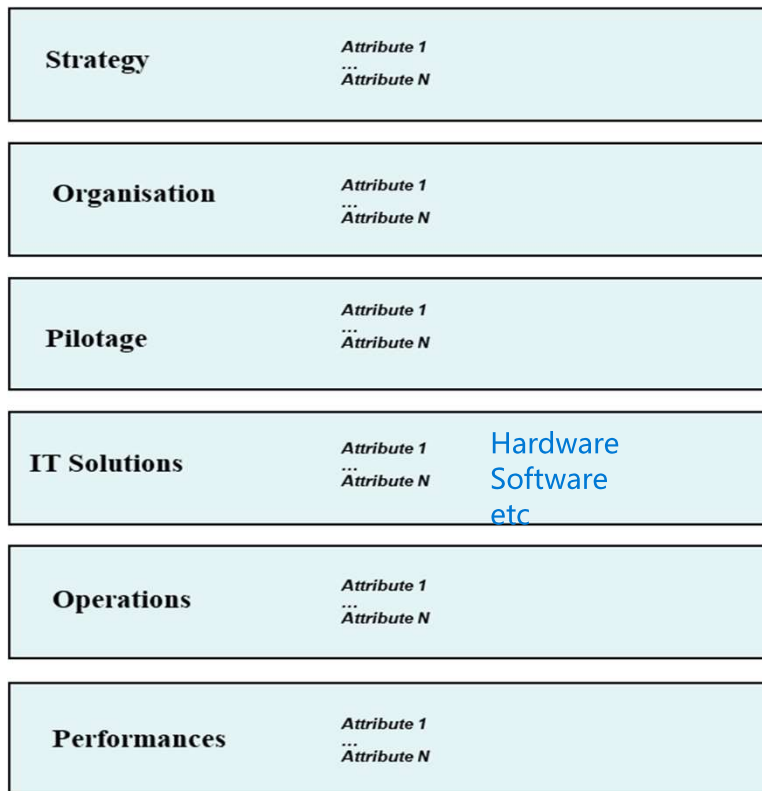
B3. DETERMINING THE LEVEL OF SUPPLY CHAIN INTEGRATION: AN EXAMPLE



DETERMINATION OF THE LEVEL OF SUPPLY CHAIN INTEGRATION

Methodology:

1. Each of the 6 fields is divided into attributes (subcomponents)



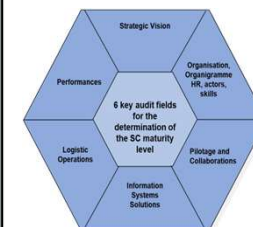
DETERMINATION OF THE LEVEL OF SUPPLY CHAIN INTEGRATION

Methodology:

1. Each of the 6 fields is divided into attributes (subcomponents)

2. Each of these attributes is evaluated with respect to the level of integration

	Level 1	Level 2	Level 3	Level 4
Strategy Attribute 1 ... Attribute N	« Silos » <i>(Clustered operations and focus per function and activity)</i>	« Integrated Company » <i>(Resources and activities are managed per groups)</i>	« Extended Company » <i>(Level 2+ Cooperation with suppliers and distributors)</i>	« Client Oriented Network » <i>(All the chains of activities are integrated and client oriented: True Supply Chain Management)</i>
Organisation Attribute 1 ... Attribute N				
Pilotage Attribute 1 ... Attribute N				
IT Solutions Attribute 1 ... Attribute N				
Operations Attribute 1 ... Attribute N				
Performances Attribute 1 ... Attribute N				



DETERMINATION OF THE LEVEL OF SUPPLY CHAIN INTEGRATION

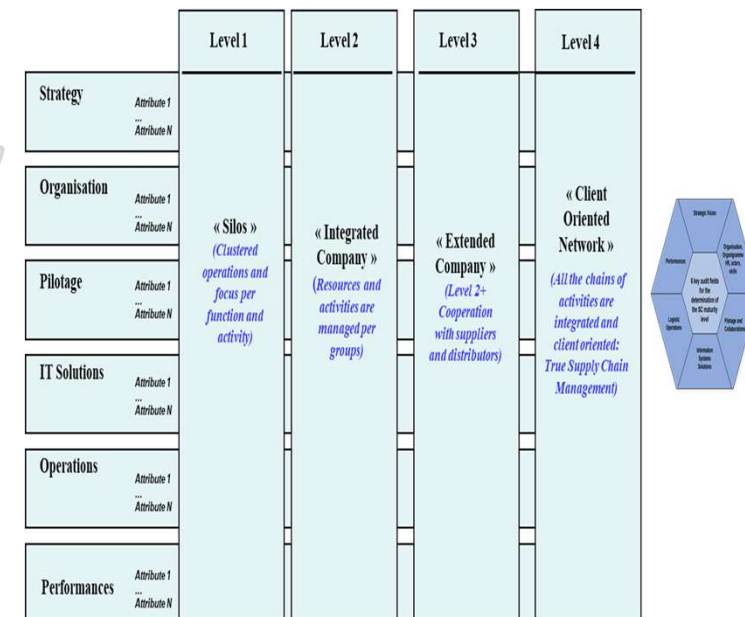
Methodology:

1. Each of the 6 fields is divided into attributes (subcomponents)
2. Each of these attributes is evaluated with respect to the level of integration

3. The consolidation (aggregation) of the above evaluations helps in highlighting, for each field, the actual level of maturity.

These are used:

- To highlight the gaps with the highest or targeted maturity level
- To initiate the search of plans and measures to lift up integration
- To benchmark with other companies' values and often the « best-in-class » companies

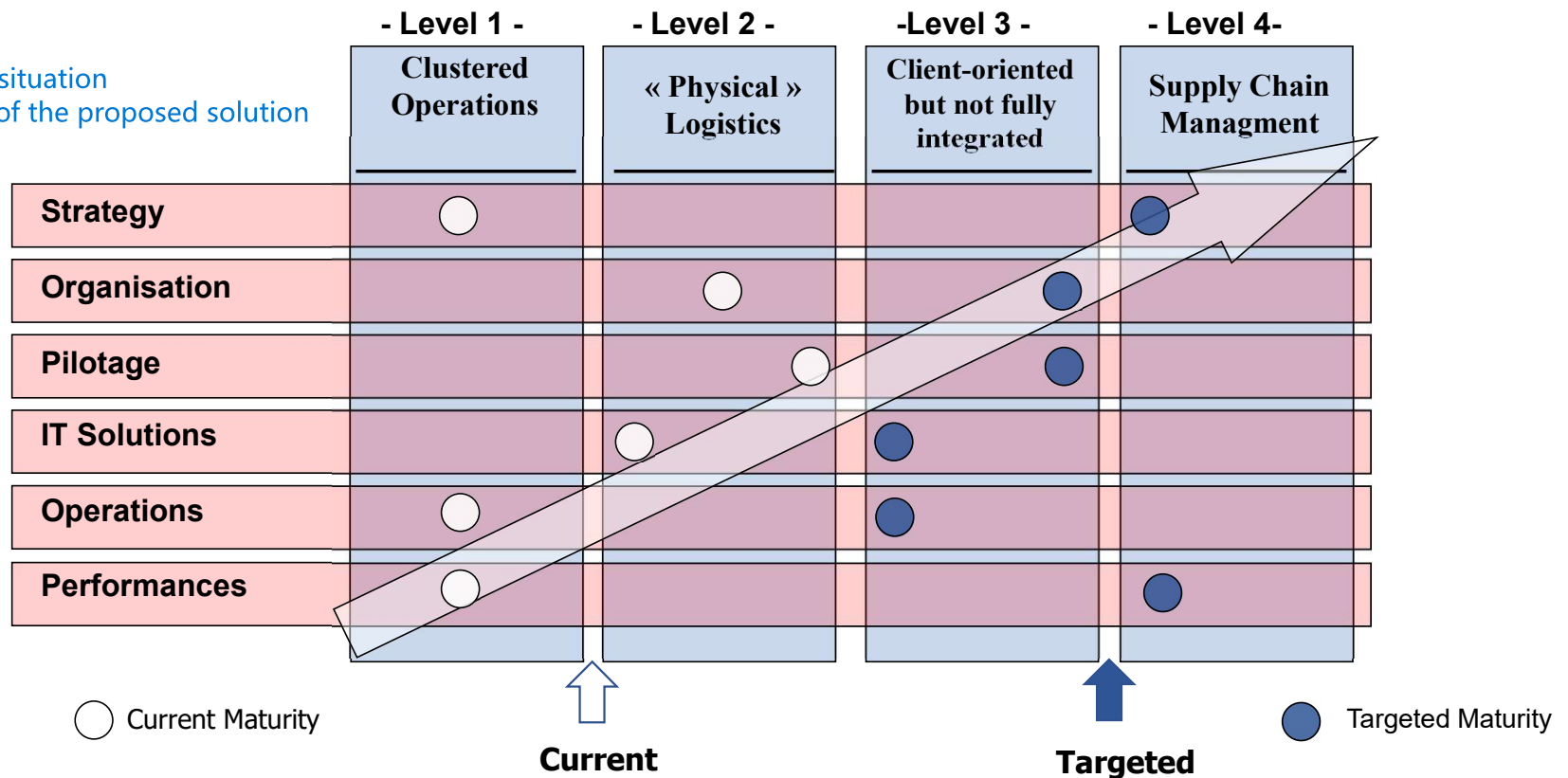


B. SUPPLY CHAIN REFERENCE MODEL AND SUPPLY CHAIN INTEGRATION

B3. DETERMINING THE LEVEL OF SUPPLY CHAIN INTEGRATION: AN EXAMPLE

Diagnostic: This company's supply chain is mostly « operations » oriented. There is no external integration, no preventive management of unexpected events (f.e. supplier failure) no integration between internal functions. The performance parameters are all centered around product delivery, product cost, product quality and not the competitive advantage.

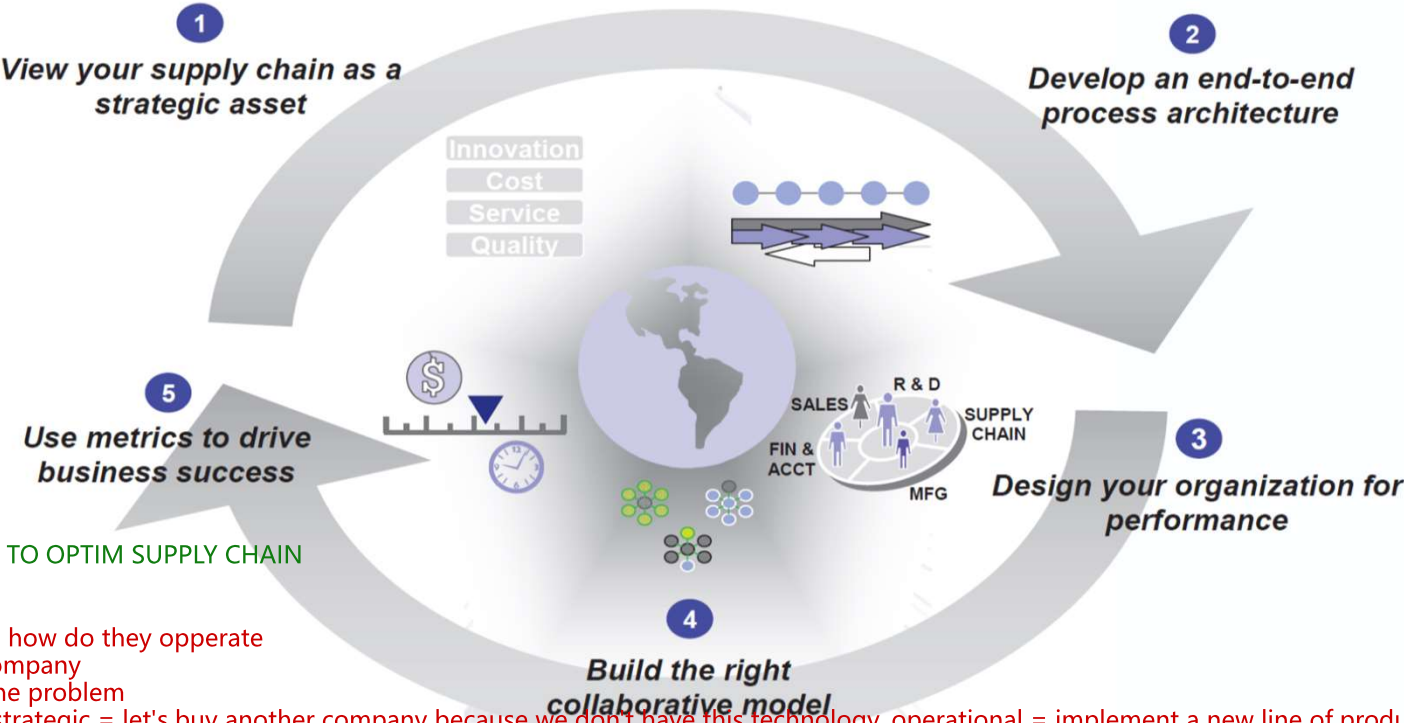
White cercles are the real situation
Blue dots is the potential of the proposed solution



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The Five Core Disciplines are the foundation for Strategic Supply Chain Management



- 1) How to describe the company, how do they operate
- 2) What is the problem of the company
- 3) What are the root causes of the problem
- 4) Generate solutions on 3 lvs (strategic = let's buy another company because we don't have this technology, operational = implement a new line of production and tactical)
- 5) Use a performance guide = the key performance indicators to track in order to be sure that the solution that I proposed works according to the plan
- 6) Tell the negative impact or what are the changes in the business that the solution will bring

Discipline 1: View Your Supply Chain as a Strategic Asset

Design the supply chain around a defined basis of competition to enable the overall business strategy

A Good Supply Chain Strategy Meets Four Basic Criteria

- 1. It is aligned with the business strategy**
 Each business strategy requires a distinct supply chain structure, processes, systems, and skills
- 2. It is aligned with customer needs**
 Understand each segment's requirements and configure the supply chain appropriately
 If necessary, create separate or multiple supply chains
- 3. It is aligned with the company's power position**
 Before making big changes, understand who has the power in the supply chain
- 4. It is adaptive**
 Supply chains have a shelf life and should be continuously evaluated to see if change is needed

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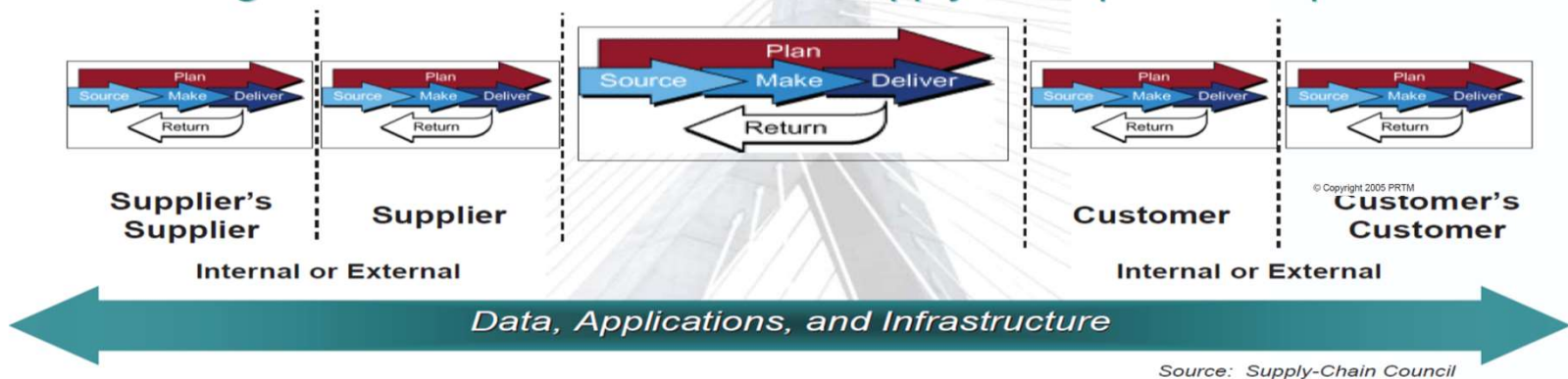
Discipline 2: Develop an End-to-End Process Architecture

Develop integrated supply chain processes and systems that interface efficiently with the rest of the enterprise

There Are Four Tests An Effective End-to-end Process Architecture Must Meet

1. Strategic Fit — Enables execution of your supply chain strategy
2. End-to-end Focus — Ensures end-to-end vision and management
3. Simplicity — Is composed of simple, streamlined processes
4. Integrity — Is highly reliable, with robust links between processes, data, and information systems

Leverage SCOR to set the context for supply chain process improvement

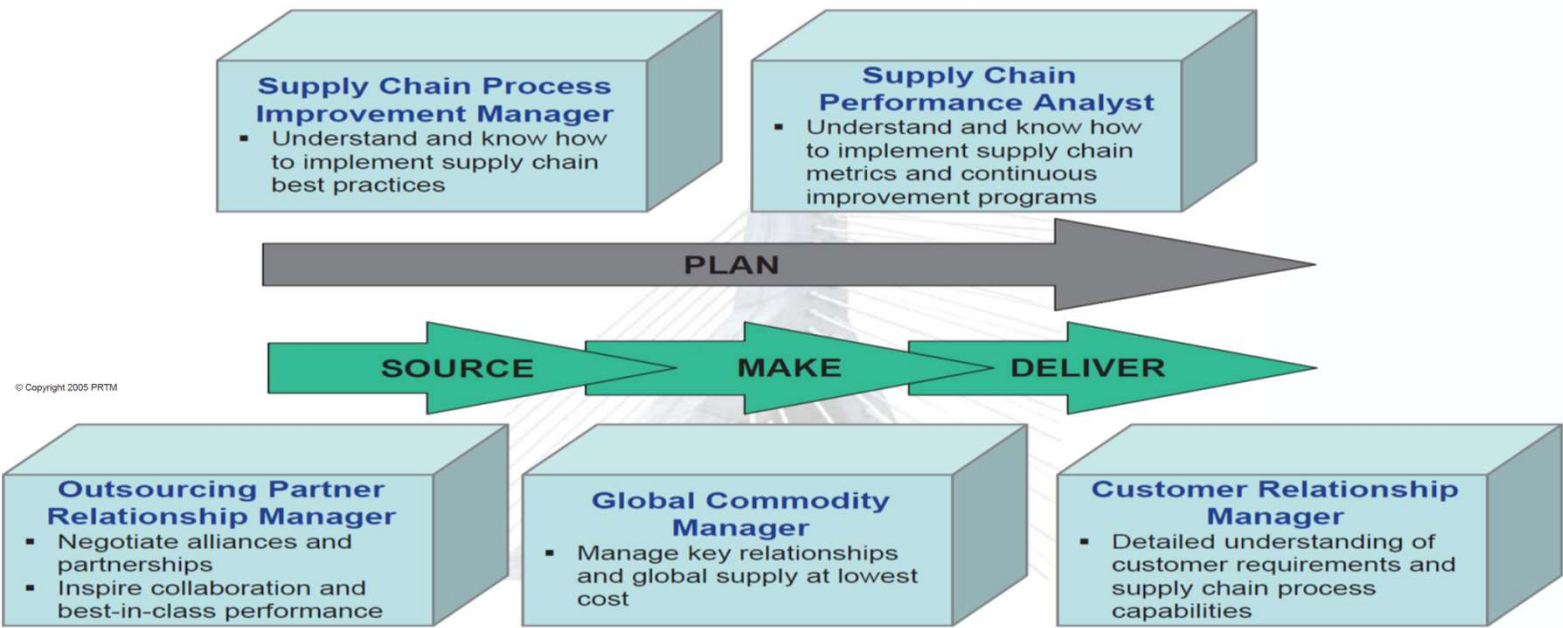


Discipline 3: Design Your Organization for Performance

Develop and maintain organizational structure and skills to define and manage the supply chain of the future

What Skills Will My Supply Chain Need In The Future?

The end-to-end supply chain will need new management skills that most companies do not possess today



Discipline 4: Built the Right Collaborative Model

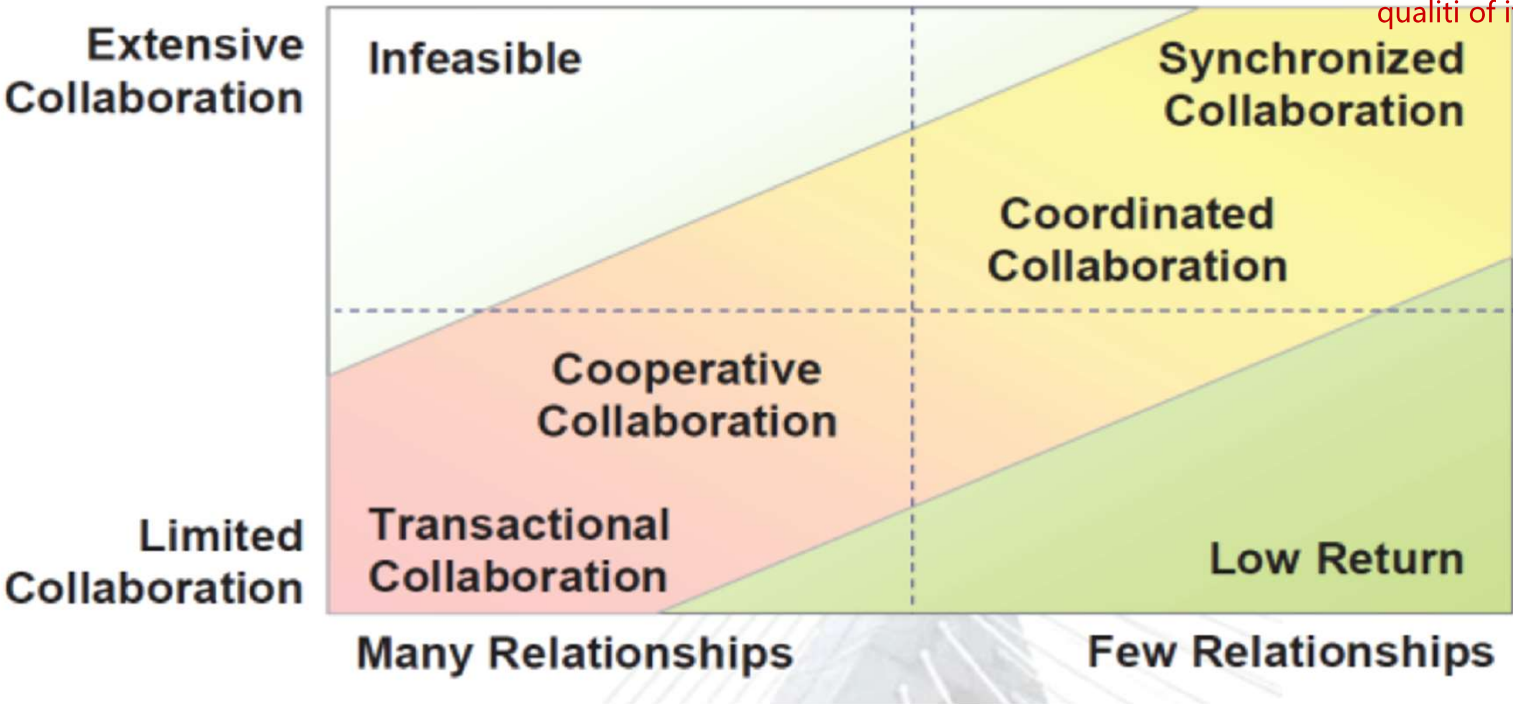
*Understand core competencies
and choose partners to maximize
focus and profitability*

FOUR BASIC MODELS OF COLLABORATIONS: NOT ONE-SIZE-FITS-ALL

We have to tell which type of collaboration do we have with the respective supplier

The Collaboration Spectrum

The supplier knows my demand in real time
And I know exactly how the supplier constructs the product and the quality of it



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Discipline 5: Use Metrics to Drive Business Success

Use metrics to measure the health of each core supply chain process and identify problem areas

SCOR Level 1 Metrics Provide A Starting Point For A Balanced Strategic Supply Chain Scorecard

Measurement is the only way to understand whether process performance is improving or worsening and whether action is required.

Key Supply Chain Management Metrics	 Delivery Performance/Responsiveness	 Flexibility	 Cost	 Asset Management
Delivery Performance	✓			
Order Fulfillment Lead Time	✓			
Production Flexibility		✓		
Total Supply Chain Management Cost			✓	
Value-Added Productivity			✓	
Inventory Days of Supply				✓
Cash-to-Cash Cycle Time				✓
Net Asset Turns				✓

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- Managing supply chain is a complex and challenging task
- The right supply chain strategy depends on a number of factors:
 - ✓ *It needs to be tailored to meet needs of the customers.*
 - ✓ *A product with stable demand and a reliable source of supply should not be managed in the same way as one with a highly unpredictable demand and an unreliable source of supply.*
- A strategy based on a "one-size-fits-all " or "try-everything " mentality will fail!
- Correct and timely information exchange is critical
- Uncertainty in Demand and Supply MUST be understood.

The Uncertainty Framework: Examples

Analyse those 2 parameters for the business case in order to propose a solution!

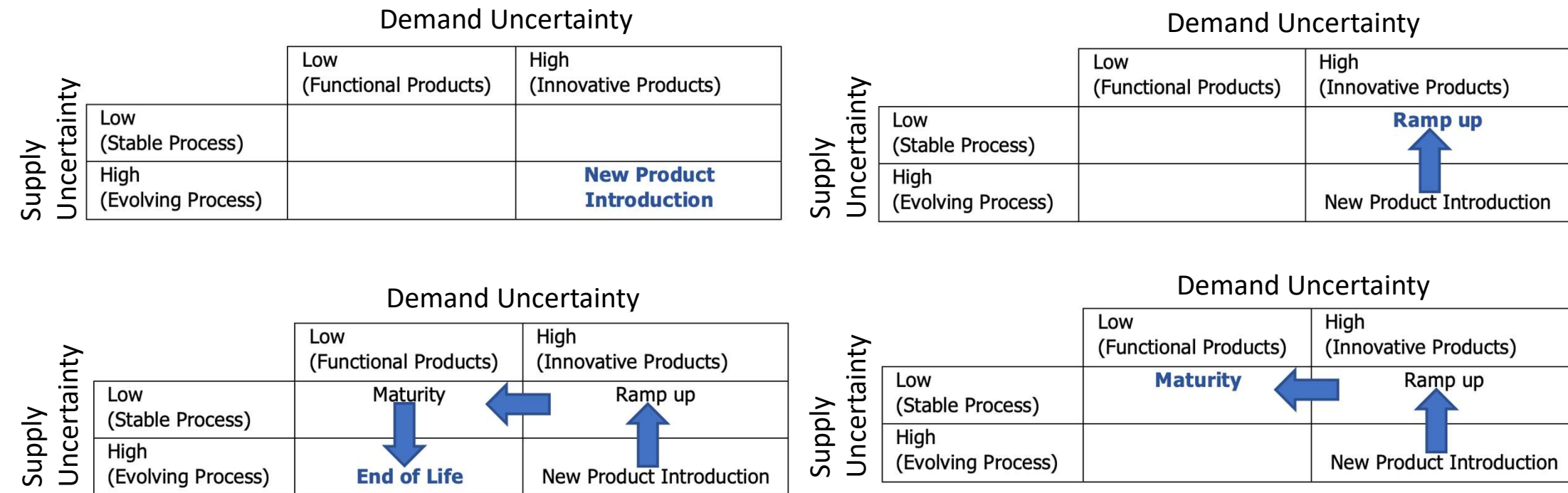
		Demand Uncertainty	
		Low (Functional Products)	High (Innovative Products)
Supply Uncertainty	Low (Stable Process)	Grocery, basic apparel, food, oil and gas	Fashion apparel, computers, pop music
	High (Evolving Process)	Hydro-electric power, some food produce	Telecom, high-end computers, semiconductor

Hau Lee, California Management Review, vol.44, n°3

When we propose a solution we have to keep this in mind, we don't want to stay in maturity for too long because we need creativity and innovation

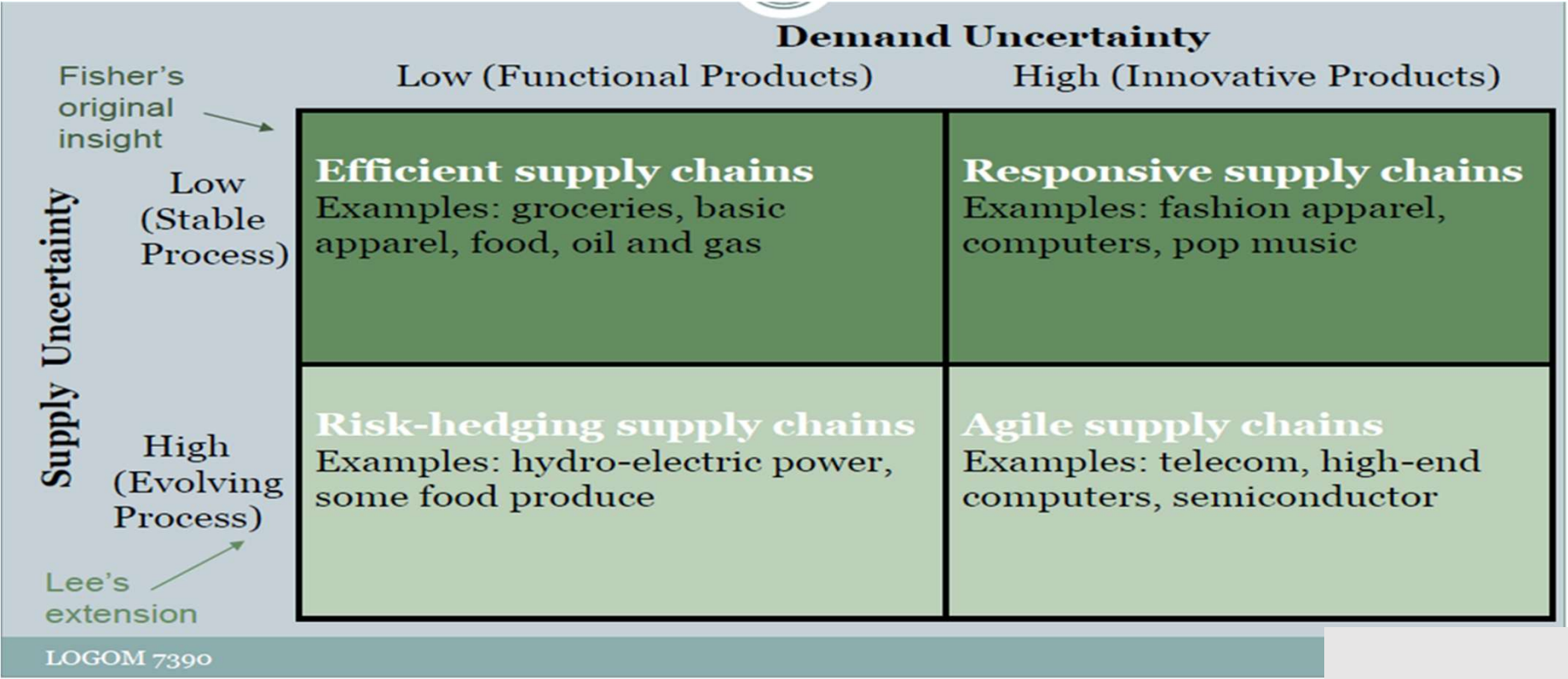
CHANGING THE UNCERTAINTY LANDSCAPE

An **adaptative strategy**: Example of Product Dynamics (*clockwise*)



Product lifecycles are now shorter than ever!!!

THE 4 GENERIC STRATEGIC SUPPLY CHAIN MANAGEMENT STRATEGIES



Based on the type of the demand that my product has, I will propose a supply strategy!!!

The most risky solution is an agile chain because we don't know if it can really support the fluctuations

C4. Focus on the **AGILE** Strategic Supply Chain Management Strategy

AGILITY should be built at every link of the
Agile Strategic Supply Chain Management Strategy :

6+1* rules of thumb:

Post ponent = Later differentiations = Ex of the color of the T-shirts

1. **Quickly Provide data on changes** in supply and demand to partners continuously so they can respond quickly.
2. **Develop collaborative relationships** with suppliers and customers so that companies work together to design or redesign processes, components, and products as well as to prepare backup plans.
3. **Introduce postponement**: Design products so that they share common parts and processes initially and differ substantially only by the end of the production process.

C4. Focus on the **AGILE** Strategic Supply Chain Management Strategy

AGILITY should be built at every link of the
Agile Strategic Supply Chain Management Strategy:

6+1* rules of thumb:

4. **Keep a small inventory of inexpensive**, nonbulky components that are often the cause of bottlenecks.
5. **Build a dependable logistics system** that can enable your company to regroup quickly in response to unexpected needs
6. **Put together a team that knows how to invoke backup plans**, contingency plans.
Intelligence team = BICC

7*. Seek additions that can strengthen the AGILE Strategic SCM Strategy

1,2,3,6 are very important points

Be sure that we are ethical, that we don't have waste, that is as green as possible, etc.

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D1. The ADAPTABILITY Addition

Objectives:

Adjust supply chain design to **meet structural shifts in markets**; modify and adapt supply network to strategies, products, and technologies.

- **Structural shifts in markets occur** because of economic progress, political and social change, demographic trends, and technological advances.
- **To stay competitive, companies must adapt their supply chains** (relocate facilities, change sources of supplies, outsource manufacturing, etc.)
- **Adaptation needn't be just a defensive tactic**
- **Companies that adapt supply chains often succeed** in launching new products or breaking into new markets.

Aligne = If we succeed as a company is because of the efforts of the different stakeholders. If we win a lot and if we share the risks with the stakeholders, we should share the gains with them as well

D. ADDITIONS TO THE BASIC STRATEGIC SUPPLY CHAIN MANAGEMENT STRATEGIES

D2. THE ALIGNMENT ADDITION

D2. The **ALIGNMENT** addition

Everyone in the chain has the same objective: to deliver the best service to consumers.

Some Alignment actions:

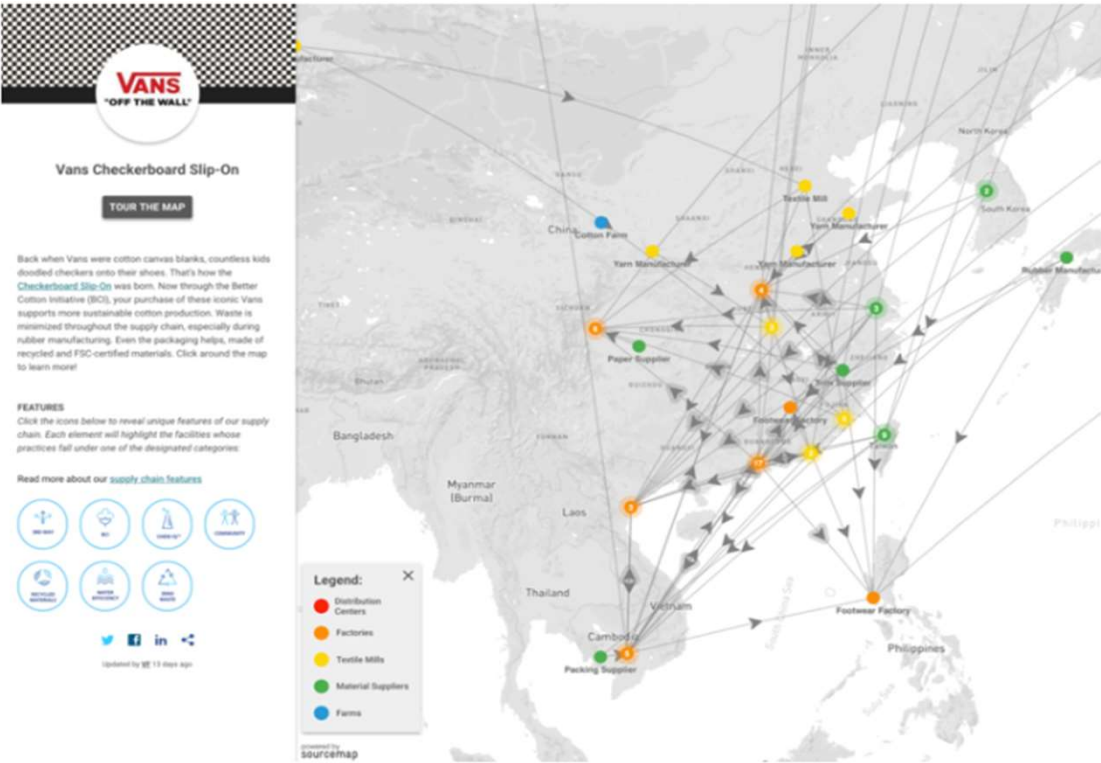
- **Redefine relationships** so that all stakeholders share risks, costs, and rewards equitably.
- **Align information**, so that all stakeholders have equal access to forecasts, sales data, and plan.
- **Define roles and responsibilities** of each partner so that there is no scope for conflict.
- **Align incentives**, so that when stakeholders maximize returns, they also maximize performance.

Requires trust and commitment on the part of suppliers, financial intermediaries, manufacturers and all other stakeholders to build competitive advantage.

D3. The CUSTOMER-FOCUSED Addition



D4. The GLOBAL addition



D5. The **FINANCIAL PROFIT-FOCUSED** Addition:

The following values MUST BE satisfactory:

- Net Profit Margin
- Gross Profit Margin
- Operating Profit Margin
- EBITDA
- Revenue Growth Rate
- Total Shareholder Return (TSR)
- Economic Value Added (EVA)
- Price/Earnings Ratio (P/E ratio)
- Return On Investment (ROI)
- Return On Capital Employed (ROCE)
- Return On Assets (ROA)
- Return On Equity (ROE)
- Debt-to-Equity (D/E) Ratio
- Cash Conversion Cycle (CCC)
- Working Capital Ratio
- Operating Expense Ratio (OER),

D6. The **ENVIRONMENTALLY CLEAN / THE GREEN** addition

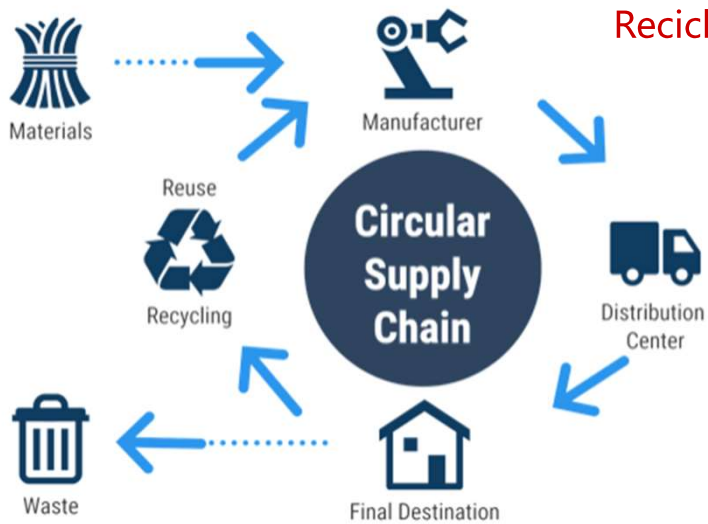
- **Green Supply Chain addition** refers to efforts to minimize the negative impact of supply chains on the natural environment.

(basically in the wake of concerns regarding climate change, pollution, and non-renewable resource constraints)

- **A Green Supply Chain focus requires** working with suppliers and customers, analysis of internal operations and processes, environmental considerations in the product development process, and extended stewardship across products' life-cycles.

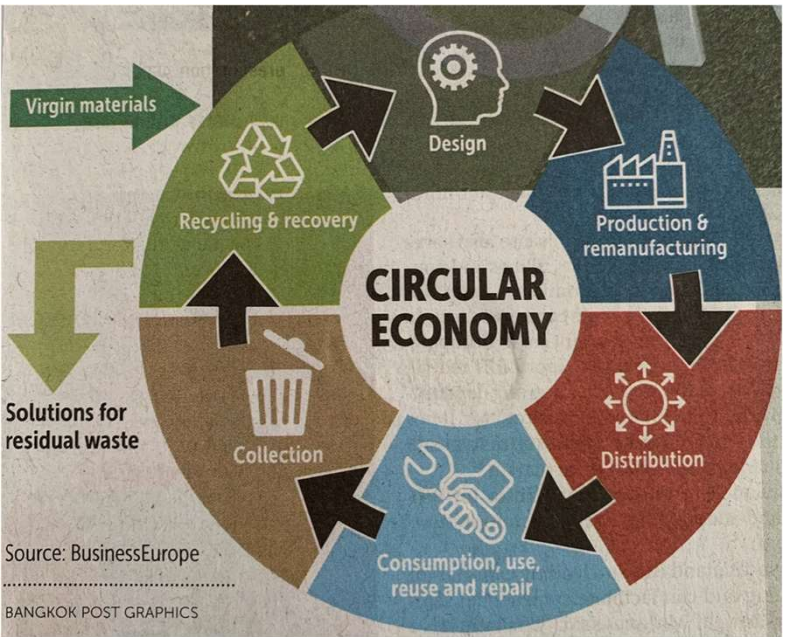
D7. The CIRCULARITY addition

Recycling the more we can to reduce the wastes



(supplychain247.com)

Looping the Supply Chain can help companies cut down spending on raw materials.



Source: BusinessEurope

BANGKOK POST GRAPHICS

Many regulations are pushing companies to adopt the circular supply chains:

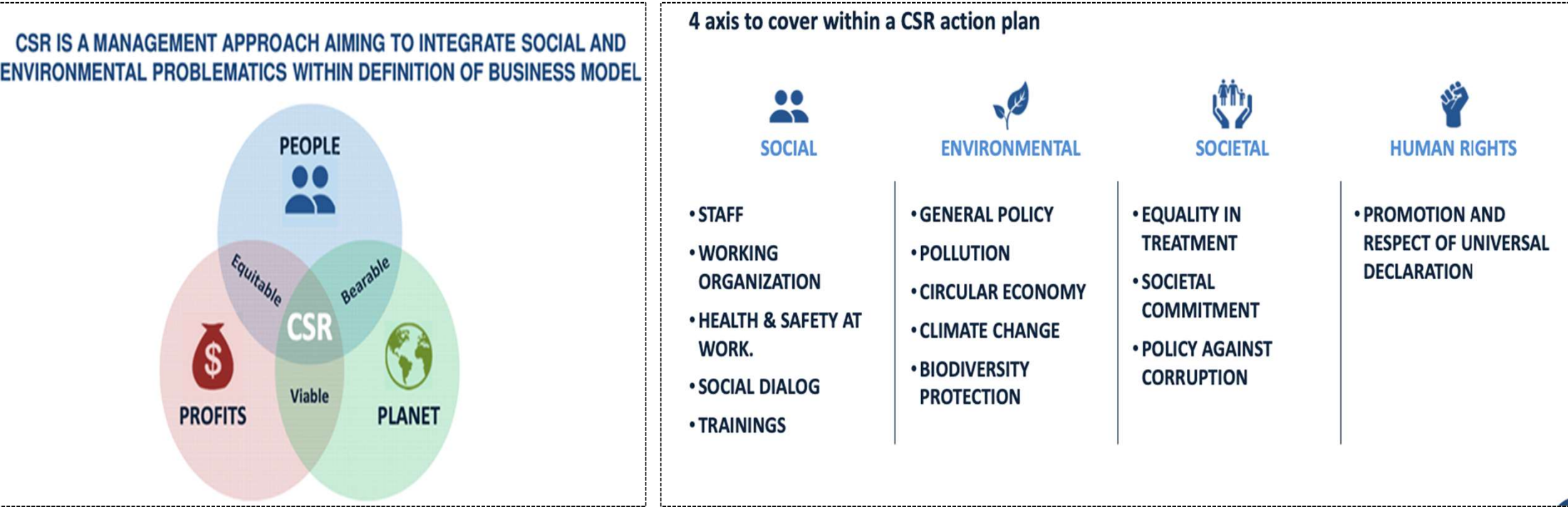
- EU Packaging Directive
- Japanese Recycling Laws
- California Recycled Content Laws
- UK Landfill Directive
- Etc.
- *New ISO14000x norms required*

D8. The SOCIALLY AND ETHICALLY RESPONSIBLE addition

right of the workers
etc.

Ultimate Objective : Contribution to societal welfare, justice and prosperity

Corporate social responsibility' (CSR) or 'Responsible Business Conduct' (RBC) is an approach used for.



D9. The **LEAN** addition

Lean addition focuses on waste reduction, helping firms eliminate non-value adding activities related to excess time, labor, equipment, space, and inventories across the supply chain.

Such strategy enables firms to improve quality, reduce costs, and improve service to customers as traditional batch and queue mass production and supply chain approaches are transformed.

Lean practices are becoming increasingly difficult to implement and sustain as supply chains increase in complexity and length.

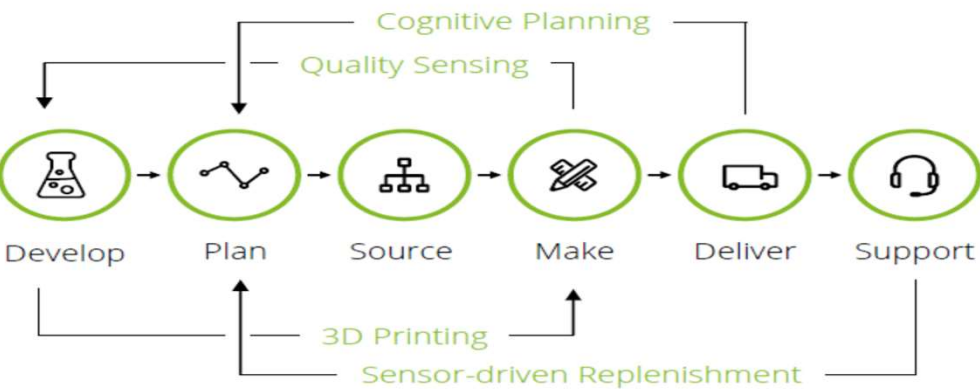
IT IS ALSO ABOUT COMPLEXITY REDUCTION!!!

D10. The SMART/DIGITAL addition

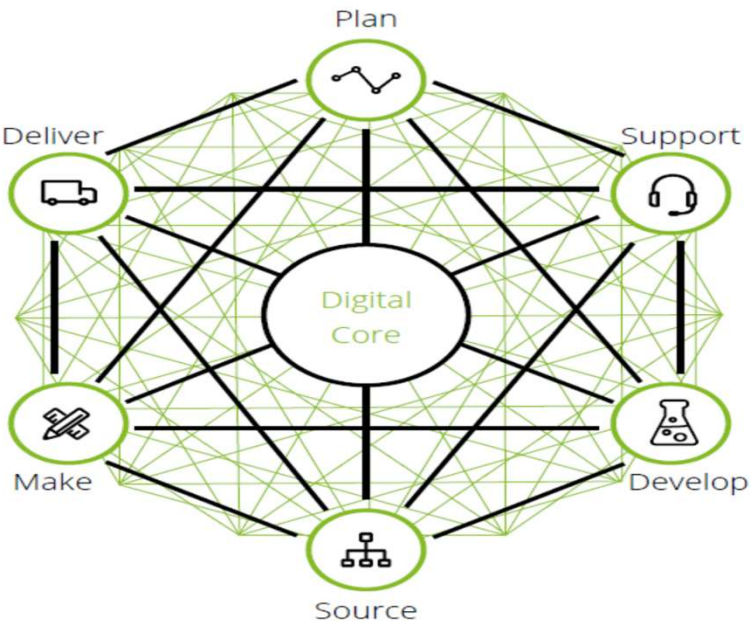
Digital Supply Networks

It transforms the traditional, linear Supply Chain nodes into a set of dynamic networks, allowing dramatically increased differentiation

Traditional Supply Chain



Digital Supply Network



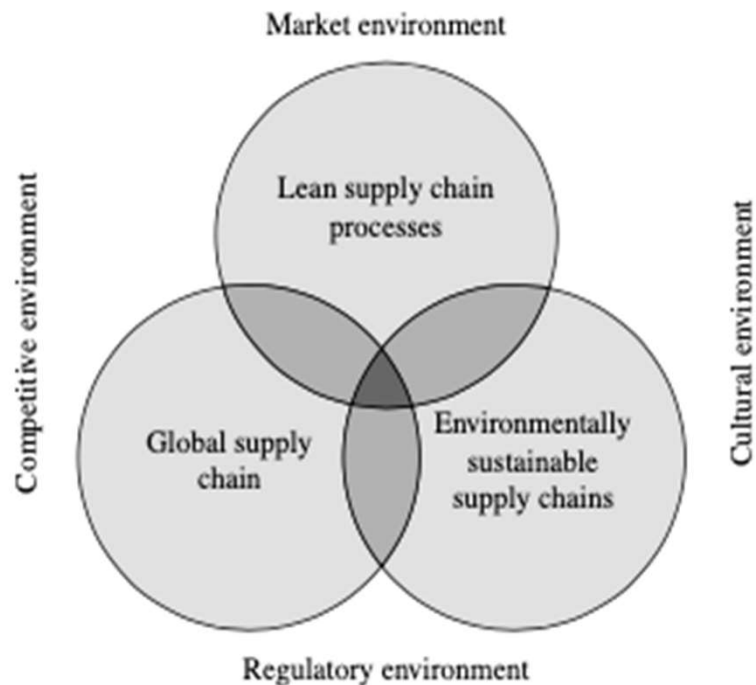
Deloitte's Point of View on Supply Chain Analytics

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 - D1. The ADAPTABILITY addition*
 - D2. The ALIGNMENT addition*
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 - D4. The GLOBAL addition*
 - D5. The FINANCIAL PROFIT-FOCUSED addition*
 - D6. The ENVIRONMENTALLY CLEAN/the GREEN addition*
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- E. ELABORATING A STRATEGIC SUPPLY CHAIN MANAGEMENT STRATEGY**
 - E1. Which Strategic Supply Chain Management Strategy is right?*
 - E2. A particular Strategic Supply Chain Management Strategy: The TRIPLE-A Strategy*
 - E3. Discussing the trade-offs paradigms in elaborating a Strategic Supply Chain Management Strategy*
- F. ASSESSING THE PERFORMANCE OF SUPPLY CHAIN MANAGEMENT STRATEGIES**
 - F1. Performance Management Science as an assessment framework*
 - F2. Building of a Supply Chain Performance Management Model*

E1. Which Strategic Supply Chain Management Strategy is right?

Joint additions or multiple parallel additions?



- **Silo perspectives impede the cross-functional scope needed to gain a holistic perspective.**
- **Stepwise simultaneous implementation makes it possible to understand the synergies available and to make appropriate trade-offs.**
- **A Systems approach helps to encapsulate the complexity and tackle supply chain strategy and decisions from a systems perspective.**

NEED TO DEVELOP SYSTEMS APPROACH AND TRADE-OFFS

Mollenkopf, Stolze, Tate, Ueltschy, (2010),
"Green, lean, and global supply chains",
IJPDL, Vol. 40 Iss: 1 pp. 14 - 41

E2. A particular combination: “The Triple-A” Strategic Supply Chain Management Strategy

Nowadays, top-performing Supply Chains possess 3 very different qualities - 3 essentials :

Agile: they react speedily to sudden change in demand or supply

Adaptable: because over time as market structures and strategies evolve

Aligned: the interests of all participants in the supply chain are aligned

This type is very much profitable over time than other supply chains types

E2. A particular combination: The “Triple-A” Strategic Supply Chain Management Strategy

Agility

Objectives:

Quickly respond to short-term changes in demand or supply; handle external disruptions smoothly.

Methods:

- Promote flow of information with suppliers and customers.
- Develop collaborative relationships with suppliers.
- Design for postponement.
- Build inventory buffers.
- Have a dependable logistics system or partner.
- Draw up contingency plans and develop crisis management teams.

Adaptability

Objectives:

Adjust supply chain’s design to meet structural shifts in markets; modify supply network to strategies, products, and technologies.

Methods:

- Monitor economies all over the world to spot new supply bases and markets.
- Use intermediaries to develop fresh suppliers and logistics infrastructure.
- Evaluate needs of ultimate consumers—not just immediate customers.
- Create flexible product designs.
- Determine where companies’ products stand in terms of technology cycles and product life cycles.

Alignment

Objective:

Create incentives for better performance.

Methods:

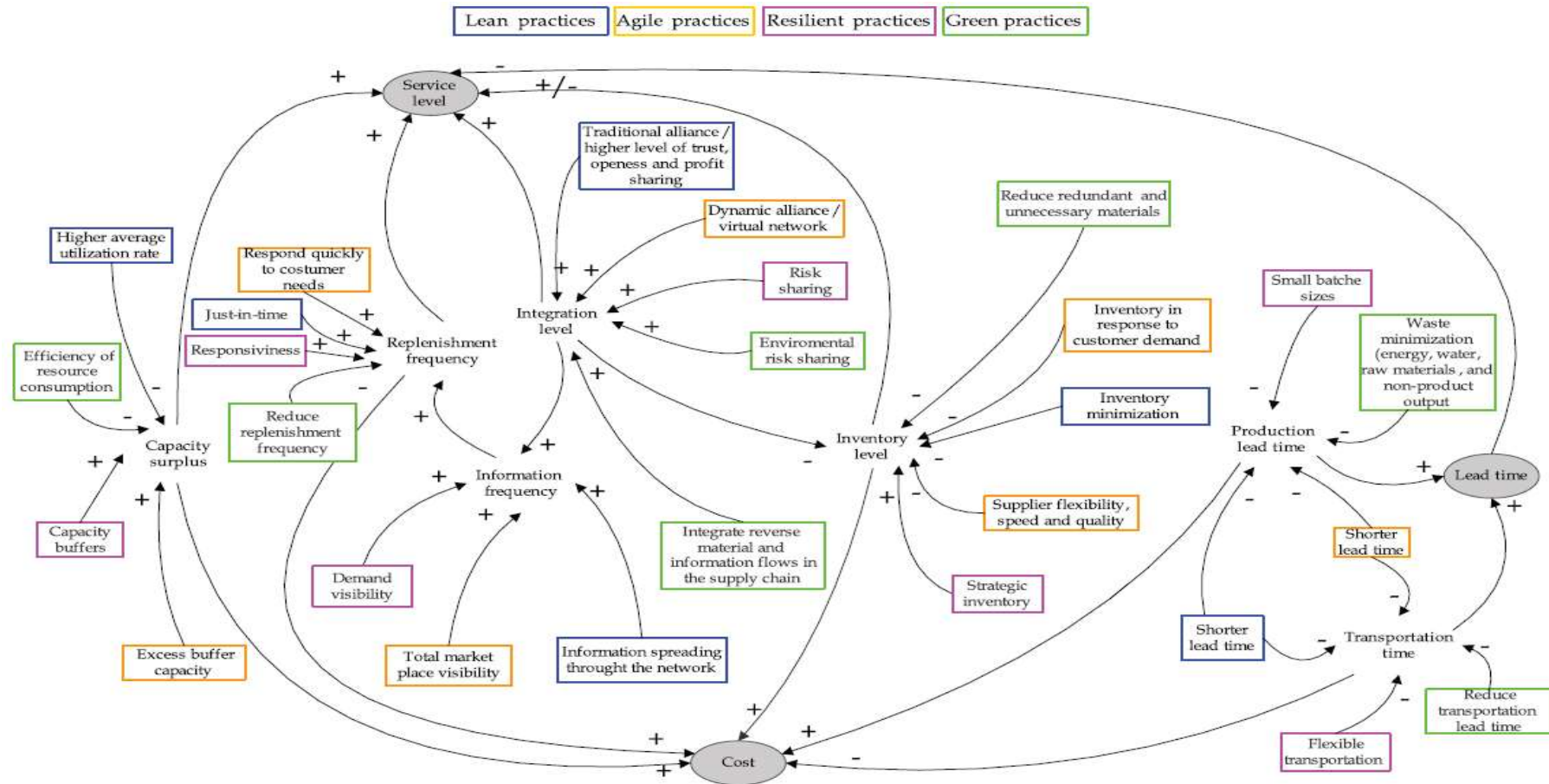
- Exchange information and knowledge freely with vendors and customers.
- Lay down roles, tasks, and responsibilities clearly for suppliers and customers.
- Equitably share risks, costs, and gains of improvement initiatives.

E3. Discussing the necessary **trade-offs** paradigms...

SOME TRADE-OFFS PARADIGMS IN SUPPLY CHAIN MANAGEMENT

- How to combine green addition with an agile basic strategy ?
- How to combine green addition with other additions when organizations are subject to disruptions and cannot be strong enough to recover competitiveness?
- How compatible are environmentally clean and global additions?
- How to ensure financial viability while staying customer-oriented and green?
- How far the smart addition is to be pushed forward?
- How to ensure a socially and ethically responsible addition while staying global?
- Etc.?

E3. Discussing the necessary trade-offs paradigms...



Conceptual model with lean, agile, resilient and green practices and supply chain Performance; Source: H. Carvalho & V. Machado

E3. Discussing the necessary **trade-offs** paradigms ...

- Building **the right supply chain strategy** is of critical urgency as globalization increases and the world faces unpredictable events brought on by climate change, geopolitical upheaval, economic uncertainty, and etc.
- The understanding of major trade-offs between additions in a holistic perspective contributes to a more efficient and competitive supply chain strategy.

But, which framework can help in making these trade-offs within a holistic perspective ?

PERFORMANCE MANAGEMENT SCIENCE offers a suitable framework for making these trade-offs possible through the assessment of supply chain management strategies.

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 - F1. *Performance Management Science as an assessment framework*
 - F2. *Building of a Supply Chain Performance Management Model*

F1. Performance Management Science as an assessment framework

PERFORMANCE MANAGEMENT

A continuous process (*not a one-shot*) that identifies opportunities for bringing improvements that can lead to industrial and business excellence through the continuous analysis of measures such as standards and indicators.



Solistica

F1. Performance Management Science as an assessment framework

PERFORMANCE MANAGEMENT

It is built around 4 pillars each containing specific approaches, tools and technologies to:

(1) Select Key Indicators and Best Standards of a business*

[Pillar 1: Key Indicators and Best Standards Selection]

(2) Gather relevant information

[Pillar 2: Data Collection and Data Cleaning]

(3) Quantify and Measure them

[Pillar 3: Performance Measurement]

(4) Report, Benchmark and Monitor them

[Pillar 4: Performance Reporting & Monitoring]



Solistica

F1. Performance Management Science as an assessment framework

PERFORMANCE MANAGEMENT

Pillar 1: Key Indicators and Best Standard Values Selection:

Indicators should provide a holistic perspective of quality, service, productivity, profitability and competitiveness.

Example of indicators: picking accuracy, shipment accuracy, proper labeling, lead time, lost sales, fill rates, throughput, asset turns, inventory turns, days on hand, revenue, return on investment, distribution excellence, operational excellence, output, staff productivity, customer satisfaction, cost of inventory, etc.

F1. Performance Management Science as an assessment framework

PERFORMANCE MANAGEMENT

Pillar 1: Key Indicators and Best Standards Selection:

Key Characteristics of indicators

- **Pertinence** (congruent with the objective, mission, goals)
- **Focus** (on accomplishable activities leading to performance)
- **Quantifiable** (from the available information/data)
- **Perennity** (stability over time)
- **Unchallengeable** by the users
- **Sensibility** (to cope with the level of detail defined)
- **Unbiased** (no possibility of multiple interpretations)
- **Accessible, easy to understand** (avoid complex indicators)
- **Convey relevant information** (for decision making)
- **Allow comparison against competitors**



F1. Performance Management Science as an assessment framework

PERFORMANCE MANAGEMENT

Pillar 1: Key Indicators and Best Standards Selection: ***Five Dimensions of indicators***

- ☐ **Dimension 1: Effectiveness**
(results obtained)
- ☐ **Dimension 2: Efficiency**
(link between decisions taken and results obtained)
- ☐ **Dimension 3: Ratio productivity/cost**
(ratio between results obtained and resources used)
- ☐ **Dimension 4: Internal targets**
(internal reference values, targets and benchmarks)
- ☐ **Dimension 5: External targets**
(external/industry reference values and benchmarks)



We can partner with competitors to exchange data or look for federal mean values to know how are we situating ourselves regarding to the competitors

F1. Performance Management Science as an assessment framework

PERFORMANCE MANAGEMENT

Pillar 1: Key Indicators and Best Standard Values Selection:

Towards KEY PERFORMANCE INDICATORS

- Some indicators roll-up and aggregate to become **Key Performance Indicators (KPI)**.
- **Key Performance Indicators (KPIs)** are process indicators and cross-functional in nature *(they measure the total effect of a process)*.
- **Example: a KPI which relates to the process "Perfect Order"**.
 - Individual processes of the "Perfect Order" are on-time, complete, correct documentation, and damage free.
 - Each individual process has its own set of indicators:
 - ✓ Indicators of on-time individual process: on-time order management, on-time order fill, on-time transportation.

F1. Performance Management Science as an assessment framework

PERFORMANCE MANAGEMENT

Pillar 1: Key Indicators and Best Standard Values Selection:
Towards KEY PERFORMANCE INDICATORS



Defining KPI trees

Deloitte, Supply Chain Analytics, 2016

Defining KPIs – KPI trees – Meta KPI – Aggregation – Decision Making

F1. Performance Management Science as an assessment framework

PERFORMANCE MANAGEMENT

Pillar 1: Key Indicators and Best Standard Values Selection: *Towards KEY PERFORMANCE INDICATORS*

KEY PERFORMANCE INDICATORS FOR LOGISTICS OPERATIONS		KEY PERFORMANCE INDICATORS FOR LOGISTICS OPERATIONS		KEY PERFORMANCE INDICATORS FOR LOGISTICS OPERATIONS	
PROCESS	INDICATOR	PROCESS	INDICATOR	PROCESS	INDICATOR
 PROCUREMENT Includes planning the demand, inventories, and purchases.	- Average cost of orders - Average time to collect - Lead time - Meeting deadlines - Percentage of errors in invoicing	 WAREHOUSE Mainly covers distribution centers and warehouses.	- Supply level - Accuracy rate for picking		- Delivery time - Average transportation cost - Cost per mile - Unplanned shipments - Cargo mix - Use of means of transportation
 MANUFACTURING Everything involving the production of goods. Also includes managing purchases, suppliers, and inventories.	- Labor cost per hour - Cost of raw materials per unit - Depreciation rate - Production level - Manufacturing time per unit - Manufacturing time per lot - Gross purchase volume - Number of suppliers - On-time deliveries	 DISTRIBUTION AND TRANSPORTATION Everything relating to moving goods from the point of origin to destination.	- Number of orders rejected - Productivity in terms of transported volume - Percentage of used space		 DELIVERY AND CUSTOMER SERVICE
				 QUALITY	- Deliveries with problems - Certified deliveries

F1. Performance Management Science as an assessment framework

PERFORMANCE MANAGEMENT

Pillar 2: Data Collection and Data Cleaning:

- Manual, semi-automatized or fully automatized data collection.
- Use of advanced IT tools and programs
- Cleaning to isolate wrong records and inconsistent data.



Solistica

F1. Performance Management Science as an assessment framework

PERFORMANCE MANAGEMENT

Pillar 3: Performance Measurement:

- Performance measurement system
 - Have all the appropriate elements (internal, external, financial, non-financial) been covered?
 - Have measures which relate to the rate of improvement been introduced?
 - Have measures which relate to the long-term and short-term objectives of the business been introduced?
 - Have the measures been integrated, both vertically and horizontally?
 - Do any of the measures conflict with any other?

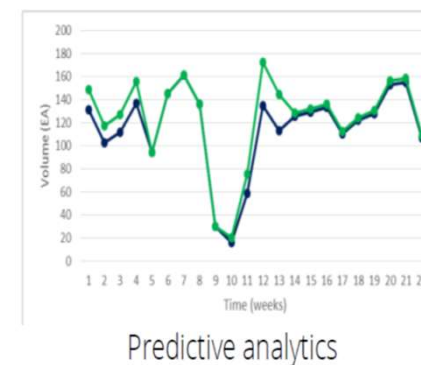
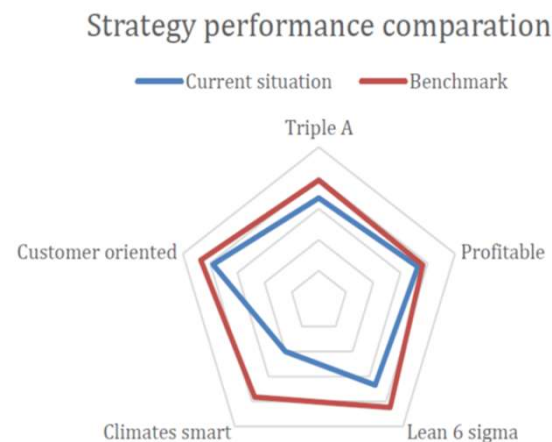
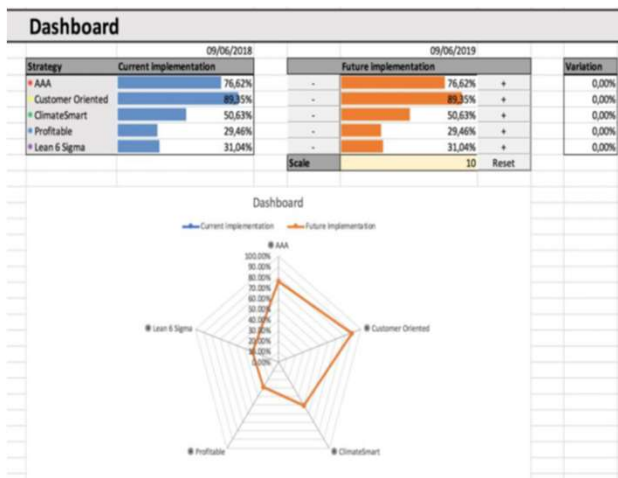


F1. Performance Management Science as an assessment framework

PERFORMANCE MANAGEMENT

Pillar 4: Performance Reporting and Monitoring:

- (1) Reporting & displaying the key indicators values using **Business Intelligence** tools
- (2) Continuously **benchmarking & monitoring** these values to allow tracking the attainment of corporate goals.



F2. Building of a Supply Chain Performance Management

OBJECTIVE

- Assess the performances of the company's supply chain strategies
- Determine the appropriateness of the company's supply chain strategies
- Highlight the potential for performance improvement, for progress
- Determine the decisions/plans to be deployed in order to reach a higher level of performance

F2. Building of a Supply Chain Performance Management Model

A. Gunasekaran et al. / Int. J. Production Economics 87 (2004) 333–347

345

Table 6
Supply chain performance metrics framework

Supply chain activity/process	Strategic	Tactical	Operational
Plan	Level of customer perceived value of product, Variances against budget, Order lead time, Information processing cost, Net profit Vs productivity ratio, Total cycle time, Total cash flow time, Product development cycle time	Customer query time, Product development cycle time, Accuracy of forecasting techniques, Planning process cycle time, Order entry methods, Human resource productivity	Order entry methods, Human resource productivity
Source		Supplier delivery performance, supplier leadtime against industry norm, supplier pricing against market, Efficiency of purchase order cycle time, Efficiency of cash flow method, Supplier booking in procedures	Efficiency of purchase order cycle time, Supplier pricing against market
Make/Assemble	Range of products and services	Percentage of defects, Cost per operation hour, Capacity utilization, Utilization of economic order quantity	Percentage of Defects, Cost per operation hour, Human resource productivity index
Deliver	Flexibility of service system to meet customer needs, Effectiveness of enterprise distribution planning schedule	Flexibility of service system to meet customer needs, Effectiveness of enterprise distribution planning schedule, Effectiveness of delivery invoice methods, Percentage of finished goods in transit, Delivery reliability performance	Quality of delivered goods, On time delivery of goods, Effectiveness of delivery invoice methods, Number of faultless delivery notes invoiced, Percentage of urgent deliveries, Information richness in carrying out delivery, Delivery reliability performance

Model's framework and indicators

- **Distinguish** the strategic, tactical and operational levels.
- **Use the SCOR framework** to cope with supply chain processes
- **Define indicators and KPIs** with clarity.

F2. Basics of Supply Chain Performance Management

Model's framework and indicators

Select indicators and group them into categories

	Measures	Indicators		Measures	Indicators		Measures	Indicators
Operational Performance	Quality	Customer reject rate	Economic Performance	Cost	New product flexibility	Environmental Performance	Green image	Number of fairs/symposiums related to environmentally conscious manufacturing the organization participate
		In plant defect fallow rate			Manufacturing cost			
		Increment products quality			Cost per operating hour			
	Customer satisfaction	After-sales service efficiency		Efficiency	Overhead expense		Business wastage	Total flow quantity of scrap
		Rates of customer complaints			Operating expenses			Percentage of materials remanufactured
		Out-of-stock ratio			Revenues from 'green' products			Percentage of materials recycled /re-used
	Delivery	On time delivery		Environmental revenues	Recycling revenues			Hazardous and toxic material output
		Delivery reliability			Cost avoidance from environmental action			Solid and liquid wastes
		Responsiveness to urgent deliveries			Cost of scrap/rework		Emissions	Energy consumption
	Time	Lead time		Environmental costs	Fines and penalties			Green house gas emissions
		Cycle times			Costs for purchasing environmentally friendly materials			Air emission
		Delivery lead time			Disposal costs			
	Inventory levels	Finished goods equivalent units			Recycling cost = transport + storage costs			
		Level of safety stocks			R & D expenses ratio			
		Order-to-ship						

F2. Building of a Supply Chain Performance Management Model

NEED TO CONSIDER THE INCREASING POWER AND SCOPE OF ANALYTICS

Competitive Advantage	High	• Optimization • Predictive Modeling • Forecasting / Extrapolation • Statistical Analysis	Predictive Analytics • What is the best that can happen? • What will happen next? • What if these trends continue? • Why is this happening?
	Low	• Alerts • Query / Drill-down • Ad-hoc Reports • Standard Reports	Descriptive Analytics • What actions are needed? • What exactly is the problem? • How many, how often, where? • What happened?
		Low	High
		Sophistication of Intelligence	

source: Accenture

F2. Building of a Supply Chain Performance Management Model

NEW METRICS AND PERFORMANCE SYSTEM

1. A new driver-based metrics framework that considers the right cross-functional and trade-offs.
2. Stable data with few unexplainable swings
3. Capable of drill-down analysis to search for root causes and to anticipate future (simulation)
4. Easily accessible to and accepted by relevant parties
5. Supported by a disciplined and documented governance process

Source: Terra Tech, University of Tennessee and Ernst & Young, 2013

PAY ATTENTION TO

- Link with Maturity Level and SCOR Matrix
- Internal (Company) and External (Competition) issues
- Benchmarking issues
- Adaptability/Transferability to other contexts
- Future updates and extensions

F2. Building of a Supply Chain Performance Management Model

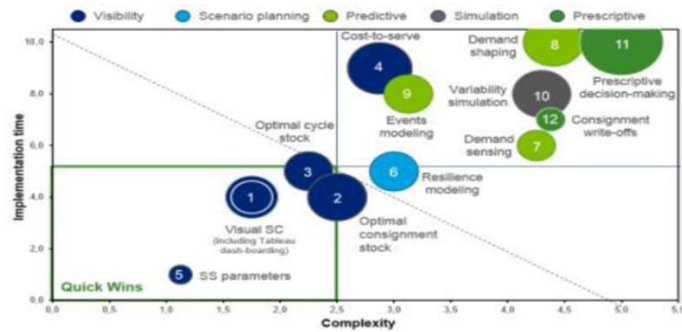
MODELS THAT PREDICT, SIMULATE, ANTICIPATE, MONITOR & HELP MANAGE



source: Ernst & Young

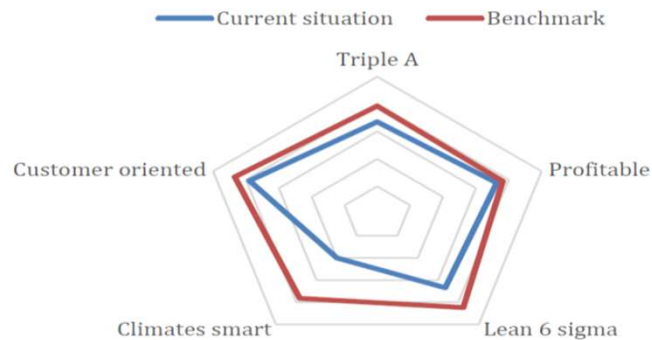
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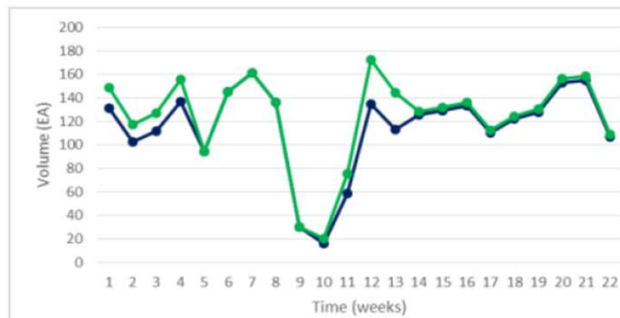


Mapping of analytics priorities

Strategy performance comparison

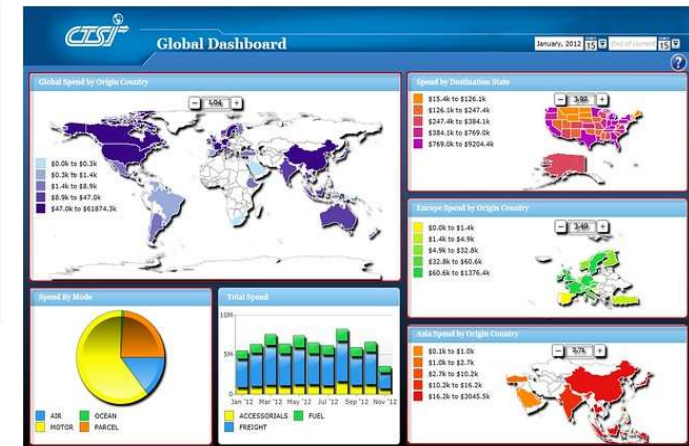
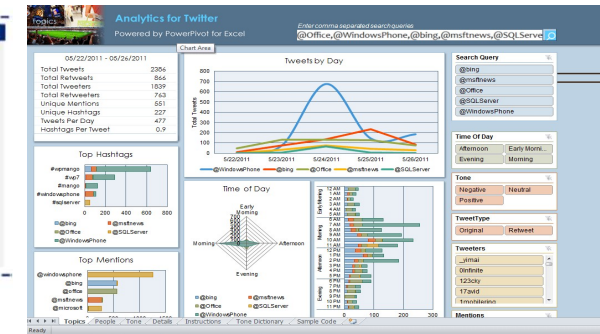


Supply Chain visualization



Predictive analytics

Deloitte's Point of View on Supply Chain Analytics



FURTHER READINGS



Northeast Supply Chain Conference



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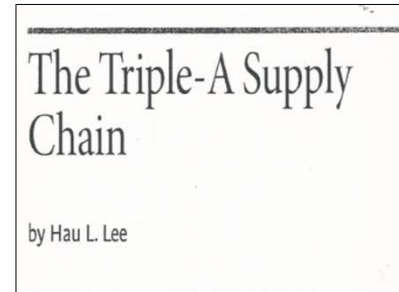
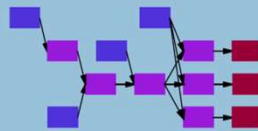


**Strategic Supply Chain
Management:
The Five Disciplines for Top
Performance**

September 20, 2005

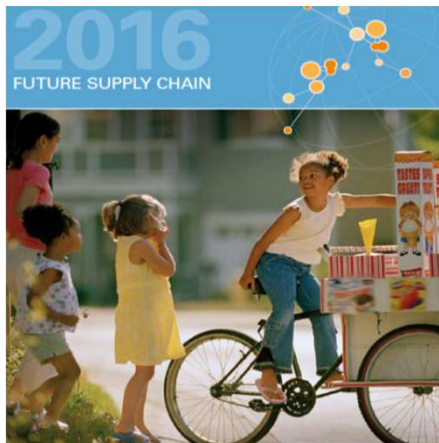
Supply Chains for the Information Age: Mastering Uncertainty and Change

M. Eric Johnson
Tuck School of Business
Dartmouth College



Chapter 2 Performance Measurement

Sara Sharahi and Maryam Abedian

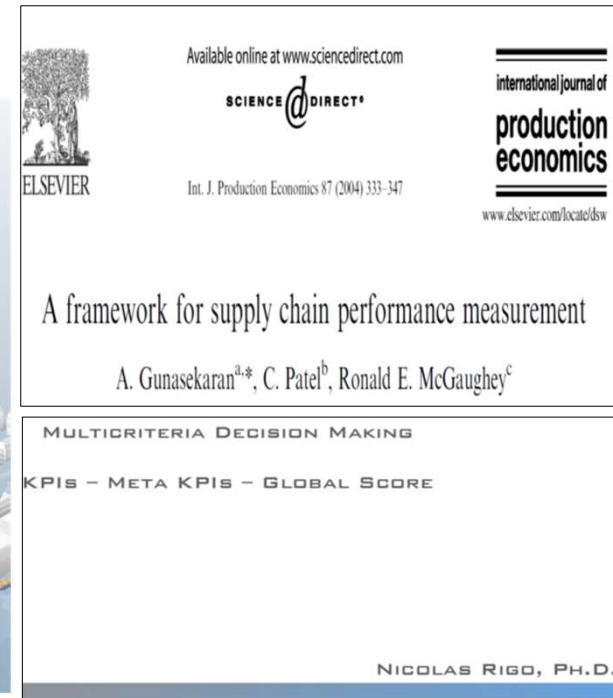


SUPPLY CHAIN WHITE PAPER
10 KEY TRENDS TO UNDERSTAND SUPPLY CHAIN MANAGEMENT

ALEXANDRE
GARNIER | APRIL
2017

AAA Supply Chains: Agility, Adaptability and Alignment

Hau L. Lee
Stanford University



Serving Consumers
in a Sustainable Way

**SUPPLY CHAIN CONCEPT
AND
SUPPLY CHAIN PERFORMANCE MANAGEMENT**

END